

Hadronic Mass from Confinement, Transition Structure and Matrix State Realization

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Abstract

This paper presents a mature hadronic mass law in which intrinsic quark mass, cumulative flavor-transition burden, and gluonic bond realization are separated into distinct physical layers. The construction begins from the two lightest realized composite anchors of the mesonic and baryonic sectors, the neutral pion and the proton, from which the mesonic bond scale $Ebase$ and baryonic bond scale $Bbase$ are extracted after subtracting explicit transition cost and intrinsic quark contributions. Their ratio defines a universal meson–baryon bond scaling factor

$$a = \frac{Bbase}{Ebase},$$

allowing a single bond law to cover both mesons and baryons.

In the mature formulation, the hadron mass is written as

$$M(H) = Estate(H) + \Delta E(H) + Q Equark,$$

so that the state sector acts only on the gluonic bond energy, while the cumulative transition term and intrinsic quark baseline remain explicitly separated. This removes the earlier ambiguity in which state acted on a blended quantity containing both bond support and transition burden.

The state-realization layer is implemented through a matrix-state architecture built from quark occupancy, flavor presence, quantum-selector, routing, activation, and ordered multiplier matrices. These matrices no longer modify the full hadron mass directly; instead, they realize the bond sector alone. In the present implementation, the revised law achieves an average absolute mass error of approximately 0.13%, with only a small number of states exceeding the 1% level.

The result is a simpler and more transparent hadronic mass framework in which the foundational layers are physically separated and the only explicitly open phenomenological layer is the state-realization map of the gluonic bond sector.

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1 Introduction

The hadronic mass problem sits between two persistent demands of theoretical physics. A successful mass law must remain physically interpretable rather than dissolve into an uncontrolled collection of fitted corrections, but it must also remain flexible enough to reproduce the wide diversity of observed mesonic and baryonic states across light, strange, charm, bottom, and excited sectors [1, 2, 3, 4, 5]. Earlier formulations of the present program already showed that a compact matrix-state description could capture a large portion of the hadronic spectrum with high numerical accuracy [6, 7]. However, they still retained one important conceptual ambiguity: the state sector acted on a prepared quantity that blended together base support and flavor-transition burden. That structure was numerically effective, but it obscured the physical role of the state layer.

The mature form developed in this paper resolves that ambiguity. Intrinsic quark mass, cumulative flavor-transition cost, and gluonic bond realization do not belong to the same physical layer and should therefore not be represented as one blended object. In the revised framework, the intrinsic quark baseline is isolated as $Q\text{ Equark}$, the cumulative flavor-transition burden is isolated as $\Delta E(H)$, and the state sector is restricted to act only on the gluonic bond energy. This yields the mature hadronic mass law

$$M(H) = Estate(H) + \Delta E(H) + Q\text{ Equark}. \quad (1)$$

The significance of this revision is not merely formal. It sharpens the interpretation of each term, prevents the state sector from compensating for explicit transition cost, and localizes the only open phenomenological layer to the realized bond structure itself.

A second key step of the mature framework is the extraction of the mesonic and baryonic bond anchors from the two simplest realized hadronic reference states: the neutral pion and the proton. Rather than introducing a free base parameter, the present paper defines the mesonic bond scale $Ebase$ and baryonic bond scale $Bbase$ by subtracting explicit quark and transition contributions from these two anchor states. Their ratio then defines a unified meson–baryon bond scaling law. In this way, the framework begins from the lightest realized hadronic composites and expands outward from them, rather than from an unexplained numerical constant [8].

The state-realization layer is implemented through an explicit matrix architecture built from quark occupancy, flavor-presence, quantum-selector, routing, activation, and ordered multiplier matrices. These matrices do not modify the full hadron mass directly. Their role is narrower and cleaner: they realize the gluonic bond sector selected by quark count and quantum configuration.

This makes the current formulation more transparent than earlier versions while preserving the compact structural logic that motivated the matrix approach in the first place [6, 7].

The broader motivation for this work is therefore twofold. First, it seeks a hadronic mass law that remains compact, explicit, and reproducible across a broad benchmark of mesons and baryons. Second, it aims to separate the foundational layers of the problem clearly enough that numerical success can be interpreted physically rather than merely recorded. That goal places the present work in continuity with the quark-model tradition inaugurated by Gell-Mann [9], but also distinguishes it from approaches in which mass prediction is recovered only through large-scale numerical machinery [4, 5]. The question pursued here is whether a comparatively economical structural law can remain stable across the observed hadronic spectrum once quark baseline, transition burden, and gluonic bond realization are assigned to their proper roles.

The paper is organized as follows. Section 2 develops the mature mathematical structure of the mass law, including the intrinsic quark baseline, the cumulative transition ladder, the extraction of the mesonic and baryonic bond anchors, and the matrix-state realization architecture. Section 3 describes the benchmark dataset and the explicit descriptor structure used in the implementation. Section 4 presents the numerical results. Section 5 discusses the physical meaning of the mature decomposition, the interpretation of the remaining tail cases, and the status of the state-realization layer as the only explicitly open phenomenological sector. Section 6 then outlines the next natural composite extension of the framework to tetraquark, pentaquark, and baryon–baryon systems, before the paper closes with its concluding remarks.

2 Mathematical Structure of the Hadronic Mass Law

This section presents the mature form of the hadronic mass law used throughout this paper. The central conceptual change relative to earlier versions is that the mass is now separated cleanly into three physically distinct layers:

1. the intrinsic quark-mass contribution,
2. the cumulative flavor-transition burden,
3. the gluonic bond sector, whose realized magnitude is determined by the state structure.

This separation removes an important ambiguity present in earlier formulations. The state sector is no longer permitted to act on a mixed quantity containing both bond support and flavor-transition cost. Instead, it acts only on the gluonic bond sector. The transition term and intrinsic quark contribution are added only after the state-resolved bond energy has been determined.

2.1 The mature universal mass law

Let H denote a hadron with Q constituent quarks. The mature mass law is

$$M(H) = \begin{cases} E_{\text{quark}} + \Delta E(H), & Q = 1, \\ E_{\text{state}}(H) + \Delta E(H) + Q E_{\text{quark}}, & Q \in \{2, 3\}. \end{cases}$$

Here:

$$\begin{aligned}
E_{\text{quark}} &: \text{intrinsic light-quark baseline,} \\
\Delta E(H) &: \text{cumulative flavor-transition burden,} \\
E_{\text{state}}(H) &: \text{state-resolved gluonic bond contribution.}
\end{aligned}$$

Thus the mass of a hadron is no longer treated as a single blended quantity. It is the sum of:
intrinsic quark baseline + explicit transition burden + state-realized gluonic bond energy.

This is the mature mathematical form of the framework.

2.2 Intrinsic quark baseline

The light-quark baseline is fixed by

$$E_{\text{quark}} = 2.17000 \text{ MeV.}$$

For a hadron with Q constituent quarks, the total intrinsic quark contribution is simply

$$Q E_{\text{quark}}.$$

This term is interpreted as the intrinsic constituent-mass layer of the hadron. It is not modified by the state sector. This is a central point of the mature framework: state does not act on the quark masses themselves. State acts only on the gluonic bond sector.

2.3 Cumulative transition burden

The flavor-transition burden is built from the ordered nearest-neighbor ladder

$$u \rightarrow d, \quad d \rightarrow s, \quad s \rightarrow c, \quad c \rightarrow b, \quad b \rightarrow t.$$

The transition constants used in the present implementation are

$$\begin{aligned}
\epsilon_{u \rightarrow d} &= 2.50, \\
\epsilon_{d \rightarrow s} &= 88.73, \\
\epsilon_{s \rightarrow c} &= 1177.60, \\
\epsilon_{c \rightarrow b} &= 2909.00, \\
\epsilon_{b \rightarrow t} &= 168320.00
\end{aligned}$$

These define the cumulative quark-elevation costs

$$\begin{aligned}
\Delta E(u) &= 0, \\
\Delta E(d) &= \epsilon_{u \rightarrow d}, \\
\Delta E(s) &= \epsilon_{u \rightarrow d} + \epsilon_{d \rightarrow s}, \\
\Delta E(c) &= \epsilon_{u \rightarrow d} + \epsilon_{d \rightarrow s} + \epsilon_{s \rightarrow c}, \\
\Delta E(b) &= \epsilon_{u \rightarrow d} + \epsilon_{d \rightarrow s} + \epsilon_{s \rightarrow c} + \epsilon_{c \rightarrow b}.
\end{aligned}$$

$$\Delta E(t) = \epsilon_{u \rightarrow d} + \epsilon_{d \rightarrow s} + \epsilon_{s \rightarrow c} + \epsilon_{c \rightarrow b} + \epsilon_{b \rightarrow t}.$$

For a hadron H with quark content $q_1, \dots, q_Q$, the total transition burden is

$$\Delta E(H) = \sum_{k=1}^Q \Delta E(q_k).$$

The mature transition law is therefore purely cumulative. There are no separate transition channels, no $L1/L2$ branching structure, and no partner-dependent transition scaling. All such logic has been removed from the transition sector and is now absorbed, where appropriate, into the state-realization layer.

2.4 Mesonic and baryonic bond anchors

The next ingredient is the gluonic bond sector. The mature framework does not begin from an abstract fitted constant. It begins from the two simplest physical anchors of the mesonic and baryonic sectors:

- the neutral pion π^0 for the mesonic bond scale,
- the proton p for the baryonic bond scale.

The mesonic bond anchor is defined by stripping the neutral pion of both its explicit transition burden and its intrinsic quark contribution:

$$E_{\text{base}} = m_{\pi^0} - \Delta E_{u \rightarrow d} - 2E_{\text{quark}}.$$

The baryonic bond anchor is defined analogously from the proton:

$$B_{\text{base}} = m_p - \Delta E_{u \rightarrow d} - 3E_{\text{quark}}.$$

These two quantities are interpreted as the residual gluonic bond support of the simplest mesonic and baryonic reference systems once quark-mass and explicit transition contributions have been removed.

This is one of the most important conceptual changes of the mature framework. The base energy is no longer treated as a free phenomenological constant. It is extracted from the lightest realized mesonic and baryonic anchor states.

2.5 Meson–baryon bond scaling

The relation between the baryonic and mesonic bond anchors is encoded in the dimensionless ratio

$$a = \frac{B_{\text{base}}}{E_{\text{base}}}.$$

This ratio is no longer an unexplained scaling parameter. It has a direct physical meaning: it is the ratio of baryonic bond support to mesonic bond support.

The bond scale used by the state sector is then written uniformly as

$$E_{\text{bond}}(Q) = E_{\text{base}} a^{Q-2}.$$

This gives immediately:

For mesons ($Q = 2$),

$$E_{\text{bond}}(2) = E_{\text{base}}.$$

For baryons ($Q = 3$),

$$E_{\text{bond}}(3) = aE_{\text{base}} = B_{\text{base}}.$$

Thus a single formula covers both the mesonic and baryonic bond sectors.

2.6 What state acts on

The mature framework imposes a sharp restriction on the state sector:

$$\boxed{\text{state acts only on } E_{\text{bond}}(Q).}$$

State does not act on $Q E_{\text{quark}}$, because intrinsic quark mass is not a state variable. State also does not act on $\Delta E(H)$, because transition cost is an explicit flavor-elevation burden rather than a bond-realization effect.

This means that the state layer is now interpreted physically as the realization map of the gluonic bond sector. Its job is to determine how the bond energy is manifested for a given quantum configuration. It is not allowed to alter the intrinsic quark masses, and it is not allowed to hide flavor-transition effects inside itself.

This is the key conceptual simplification of the mature mass law.

2.7 State-realization functional

The state-resolved bond contribution is written as

$$\boxed{E_{\text{state}}(H) = \chi(E_{\text{bond}}(Q); s, I, I_3, \sigma_{sd}, G, Q, Q_u, Q_d, Q_s, Q_c, Q_b, N_s, N_c, N_b, \ell, n),}$$

where χ denotes the state-realization functional.

Its inputs are:

- s : spin,
- I : isospin,
- I_3 : isospin projection,
- σ_{sd} : spin–diquark symmetry indicator,
- G : highest generation present in the hadron,
- Q : number of constituent quarks,
- Q_u, Q_d, Q_s, Q_c, Q_b : flavor-presence selectors,
- N_s, N_c, N_b : heavy-flavor multiplicities,
- ℓ : orbital label,
- n : radial/state label.

In code, this is implemented as the function

$$\text{stateEnergy}(\cdots),$$

which returns the realized bond energy for the configuration in question.

The mature functional form remains state dependent. Different state levels have different realization rules. This is intentional. The foundational layers of the equation are now first-principles, while the state-realization matrix remains the one explicitly open phenomenological layer of the framework.

2.8 Generation structure

The quark generations used in the present construction are

$$\boxed{u, d \in G_1, \quad s, c \in G_2, \quad b, t \in G_3.}$$

The generation indicator

$$G = \max\{G(q_1), \dots, G(q_Q)\}$$

is used as an input selector inside the state-realization functional. In the mature formulation, generation no longer drives separate transition channels. Instead, it helps determine which state-realization rules are active once the bond sector is isolated.

2.9 Flavor presence and multiplicity selectors

The state-realization layer also uses flavor-presence selectors

$$Q_u, Q_d, Q_s, Q_c, Q_b \in \{0, 1\},$$

which indicate whether a given flavor is present at least once, and multiplicity counters

$$N_s, N_c, N_b \in \{0, 1, 2, 3\},$$

which count repeated heavy-flavor occupancy.

These serve two purposes.

First, they allow the state-functional to distinguish between qualitatively different composition classes such as

$$(u, d), \quad (u, s), \quad (c, s), \quad (c, c), \quad (b, b),$$

or

$$(u, d, s), \quad (u, s, c), \quad (s, s, c), \quad (u, d, b),$$

without reintroducing transition-channel branching.

Second, they localize the remaining open structure of the framework to the state layer, where configuration genuinely matters.

2.10 Mesonic and baryonic state classes

The state functional is split first by hadron size:

$$Q = 2 \quad (\text{mesons}), \quad Q = 3 \quad (\text{baryons}),$$

and then by generation class G . Within each such sector, the realized bond energy is obtained by multiplying the bond anchor $E_{\text{bond}}(Q)$ by a state-specific set of selector powers.

Schematically, for a given state n ,

$$E_{\text{state}}(H) = E_{\text{bond}}(Q) \mathcal{M}_n(H),$$

where $\mathcal{M}_n(H)$ is a product of factors activated by the hadron's quantum descriptors. The explicit factors differ from state to state, and the mature implementation preserves those distinctions rather than forcing all states into one universal multiplier.

This is important. The mature framework simplifies the mass law globally, but it does not erase the local structure of the realized-state matrix. The detailed state rules remain essential.

2.11 Single-quark sector

For completeness, the single-quark sector is treated separately:

$$M(q) = E_{\text{quark}} + \Delta E(q).$$

No gluonic bond or state-realization term is used in this case, since the bond sector is a composite-sector phenomenon. This gives the exact baseline quark masses used throughout the model.

2.12 Final output structure

Combining all layers, the mature hadronic mass law may be summarized as follows.

For a quark:

$$M(q) = E_{\text{quark}} + \Delta E(q).$$

For a meson or baryon:

$$M(H) = E_{\text{state}}(H) + \Delta E(H) + Q E_{\text{quark}},$$

with

$$E_{\text{state}}(H) = \chi(E_{\text{base}} a^{Q-2}; \text{quantum descriptors of } H).$$

Thus the mature logic flow is:

intrinsic quark mass + cumulative flavor-transition burden + state-realized gluonic bond energy.

This is the final mathematical structure of the mass law used in the present paper.

2.13 Interpretive summary

The mature form of the framework has three major advantages.

First, it begins from physical anchors rather than unexplained constants. The mesonic and baryonic bond scales are extracted from the π^0 and the proton.

Second, it isolates the three physical layers of the hadron mass:

- intrinsic quark baseline,
- explicit flavor-transition burden,
- state-realized gluonic bond.

Third, it localizes the only open phenomenological structure to the state-realization layer. Everything else is now grounded directly in the quark baseline, the transition ladder, and the bond anchors.

This sharpened separation is what makes the mature hadronic mass law both conceptually cleaner and numerically stronger than its earlier versions.

2.14 Parent quark occupancy and presence matrices

The parent compositional object of the hadron is the quark-occupancy matrix

$$\hat{N}_q(H) = \text{diag}(N_u(H), N_d(H), N_s(H), N_c(H), N_b(H)),$$

where each entry satisfies

$$N_f(H) \in \{0, 1, 2, 3\}.$$

Thus, for example,

$$\hat{N}_q(H) = \begin{pmatrix} 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

represents a same-flavor uu meson sector, while

$$\hat{N}_q(H) = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

represents a dsc -type baryon.

From $\hat{N}_q(H)$ one derives the quark-presence matrix

$$\hat{P}_q(H) = \text{diag}(Q_u(H), Q_d(H), Q_s(H), Q_c(H), Q_b(H)),$$

where

$$Q_f(H) = \begin{cases} 1, & N_f(H) \geq 1, \\ 0, & N_f(H) = 0. \end{cases}$$

The occupancy matrix records multiplicity. The presence matrix records whether a flavor is present at all. Both are needed. Occupancy distinguishes, for example, ss from us ; presence distinguishes broad flavor classes without forcing the state rules to work directly at the raw occupancy level.

In the mature formulation, these matrices no longer feed a partner-dependent transition machine. Instead, they act as binary and multiplicity inputs for the state-realization layer. The transition burden has already been isolated separately in $\Delta E(H)$; the purpose of $\hat{N}_q(H)$ and $\hat{P}_q(H)$ is to determine which bond-realization rules may fire.

2.15 Quantum selector matrices

The state-realization functional depends on the hadron's quantum descriptors. These are encoded through selector matrices.

Spin selector matrix. Using the spin basis appropriate to the hadronic dataset,

$$B_s = \left\{ 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}, 3 \right\},$$

define

$$\hat{P}_s(H) = \text{diag}(p_{s=0}, p_{s=1/2}, p_{s=1}, p_{s=3/2}, p_{s=2}, p_{s=5/2}, p_{s=3}),$$

where

$$p_{s=\sigma} = \begin{cases} 1, & s(H) = \sigma, \\ 0, & s(H) \neq \sigma. \end{cases}$$

Isospin selector matrix. Using the isospin basis

$$B_I = \left\{ 0, \frac{1}{2}, 1, \frac{3}{2} \right\},$$

define

$$\hat{P}_I(H) = \text{diag}(p_{I=0}, p_{I=1/2}, p_{I=1}, p_{I=3/2}).$$

Absolute- I_3 selector matrix. Because a number of state rules depend on $|I_3|$ rather than on its sign, define

$$B_{|I_3|} = \left\{ 0, \frac{1}{2}, 1, \frac{3}{2} \right\},$$

and

$$\hat{P}_{|I_3|}(H) = \text{diag}(p_{|I_3|=0}, p_{|I_3|=1/2}, p_{|I_3|=1}, p_{|I_3|=3/2}).$$

Orbital selector matrix. Using the orbital basis

$$B_\ell = \{0, 1\},$$

define

$$\hat{P}_\ell(H) = \text{diag}(p_{\ell=0}, p_{\ell=1}).$$

sd-symmetry selector matrix. Using the symmetry basis

$$B_\sigma = \{0, 1\},$$

define

$$\hat{P}_\sigma(H) = \text{diag}(p_{\sigma=0}, p_{\sigma=1}).$$

These selector matrices are binary. Their role is not to generate energy directly. Their role is to determine which entries of the state-realization matrix are activated for the bond sector

$$E_{\text{bond}}(Q) = E_{\text{base}} a^{Q-2}.$$

2.16 Routing matrices and block selection

The mature state machine is routed first by quark count, then by generation in the meson sector, and finally by state label.

Quark-count routing. Using the basis

$$B_Q = \{1, 2, 3\},$$

define

$$\hat{P}_Q(H) = \text{diag}(p_{Q=1}, p_{Q=2}, p_{Q=3}).$$

This distinguishes:

$$Q = 1 \quad (\text{single quark}), \quad Q = 2 \quad (\text{meson}), \quad Q = 3 \quad (\text{baryon}).$$

Meson-generation routing. For mesons, generation acts as a rule-enabling mask. Using the basis

$$B_G = \{1, 2, 3\},$$

define

$$\hat{P}_G(H) = \text{diag}(p_{G=1}, p_{G=2}, p_{G=3}).$$

Here G is the highest generation present in the hadron:

$$G(H) = \max\{G(q_1), \dots, G(q_Q)\}.$$

State routing for mesons. Using the mesonic state basis

$$B_n^{(2)} = \{0, 1, 2, 3, 4, 5, 6\},$$

define

$$\hat{P}_n^{(2)}(H) = \text{diag}(p_{n=0}, p_{n=1}, p_{n=2}, p_{n=3}, p_{n=4}, p_{n=5}, p_{n=6}).$$

State routing for baryons. Using the baryonic state basis

$$B_n^{(3)} = \{1, 2, 3, 4, 5\},$$

define

$$\hat{P}_n^{(3)}(H) = \text{diag}(p_{n=1}, p_{n=2}, p_{n=3}, p_{n=4}, p_{n=5}).$$

Mesons therefore route by the ordered triple

$$(Q, G, n),$$

whereas baryons route by

$$(Q, n),$$

with heavy/light and flavor-content distinctions handled internally by the compositional selectors derived from $\hat{N}_q(H)$ and $\hat{P}_q(H)$.

This routing structure is unchanged in spirit from earlier versions, but its target has changed: it now selects the realized gluonic bond state rather than acting on a blended confinement-plus-transition object.

2.17 State-rule activation matrices

For each active block, define the ordered diagonal activation matrix

$$\hat{A}^{(Q,G,n)}(H) = \text{diag}(a_1(H), a_2(H), \dots, a_m(H)), \quad a_j(H) \in \{0, 1\}.$$

Each activation $a_j(H)$ is a product of binary selector readouts from the parent matrices. Because all selector outputs are binary, multiplication functions as logical AND.

For example,

$$a_j(H) = p_{s=1} p_{I=1/2} Q_c(H)$$

fires if and only if the hadron simultaneously satisfies

$$s = 1, \quad I = \frac{1}{2}, \quad Q_c = 1.$$

Likewise,

$$a_j(H) = p_{s=0} p_{I=1} p_{|I_3|=1}$$

fires if and only if

$$s = 0, \quad I = 1, \quad |I_3| = 1.$$

And

$$a_j(H) = p_{s=1} p_{\ell=1} \mathbf{1}(N_c(H) = 2)$$

fires if and only if

$$s = 1, \quad \ell = 1, \quad N_c = 2.$$

In the mature formulation, the activation matrices are interpreted as selectors of admissible bond-realization channels. They do not act on the explicit transition burden $\Delta E(H)$, and they do not modify the intrinsic quark baseline QE_{quark} . Their sole role is to determine which ordered multiplier entries act on the isolated bond scale $E_{\text{bond}}(Q)$.

2.18 Ordered multiplier matrices

The diagonal multiplier matrices are frozen in the exact order of the implemented rules in the current code. The activation matrices must use the same order. In the mature formulation, every multiplier listed below acts on

$$E_{\text{bond}}(Q) = E_{\text{base}} a^{Q-2},$$

and never on the full quantity

$$\Delta E(H) + QE_{\text{quark}}.$$

Thus, if $\hat{\Lambda}^{(Q,G,n)}$ denotes the ordered multiplier matrix for a given block, the realized bond contribution is obtained schematically as

$$E_{\text{state}}(H) = E_{\text{bond}}(Q) \mathcal{M}(H),$$

where $\mathcal{M}(H)$ is the product of the active multiplier entries selected by the binary activation matrix $\hat{A}^{(Q,G,n)}(H)$.

Equivalently, one may write

$$\mathcal{M}(H) = \prod_{j=1}^m \lambda_j^{a_j(H)},$$

where λ_j are the ordered diagonal entries of the relevant multiplier matrix. Since $a_j(H) \in \{0, 1\}$, inactive channels contribute multiplicative identity, and active channels contribute their realized multiplier.

This reproduces the implemented rule logic exactly.

Mesons: $Q = 2$, $G = 1$. For $n = 0$,

$$\hat{\Lambda}^{(2,1,0)} = (7.5947).$$

For $n = 1$,

$$\hat{\Lambda}^{(2,1,1)} = \text{diag}(4.2222, 1.0358, 5.9957, 10.2340).$$

For $n = 2$,

$$\hat{\Lambda}^{(2,1,2)} = \text{diag}(11.4097, 10.0452, 10.9505, 9.9008).$$

For $n = 3$,

$$\hat{\Lambda}^{(2,1,3)} = \text{diag}(13.6117, 14.0878, 12.9795, 13.3697, 13.1263).$$

Mesons: $Q = 2$, $G = 2$. For $n = 1$,

$$\hat{\Lambda}^{(2,2,1)} = \text{diag}(1.3037, 3.6152, 3.1180, 4.6275, 5.8359, 6.4982, 6.2129, 5.7257, 4.6152, 3.4486, 4.3305, 6.8108, 7.6744, 7.9147).$$

For $n = 2$,

$$\hat{\Lambda}^{(2,2,2)} = \text{diag}(7.4404, 8.1540, 8.7978, 8.8363, 11.6530, 9.42899, 8.9770, 9.2825, 1.1490, 8.9287).$$

For $n = 3$,

$$\hat{\Lambda}^{(2,2,3)} = \text{diag}(13.0111, 9.5725, 10.5638, 10.57332, 9.6467, 9.9909, 10.1508).$$

For $n = 4$,

$$\hat{\Lambda}^{(2,2,4)} = \text{diag}(15.8752, 11.6308, 11.6691, 11.6828).$$

For $n = 5$,

$$\hat{\Lambda}^{(2,2,5)} = \text{diag}(13.1078, 12.8690).$$

For $n = 6$,

$$\hat{\Lambda}^{(2,2,6)} = (14.6640).$$

Mesons: $Q = 2$, $G = 3$. For $n = 1$,

$$\hat{\Lambda}^{(2,3,1)} = \text{diag}(6.4298, 8.5337, 8.1085, 8.5606, 8.9123, 8.5869, 8.9165, 11.9622).$$

For $n = 2$,

$$\hat{\Lambda}^{(2,3,2)} = \text{diag}(12.7910, 14.6133, 12.9803, 14.8260, 14.8958, 12.0388, 12.1798, 11.1131).$$

For $n = 3$,

$$\hat{\Lambda}^{(2,3,3)} = \text{diag}(16.0882, 13.0958, 12.3040).$$

For $n = 4$,

$$\hat{\Lambda}^{(2,3,4)} = \text{diag}(13.8823, 13.9408, 13.9663, 1.0285, 17.3206).$$

For $n = 5$,

$$\hat{\Lambda}^{(2,3,5)} = (19.7055).$$

For $n = 6$,

$$\hat{\Lambda}^{(2,3,6)} = (20.6030).$$

Baryons: $Q = 3$. The baryonic blocks act on

$$E_{\text{bond}}(3) = B_{\text{base}} = aE_{\text{base}}.$$

For $n = 1$,

$$\hat{\Lambda}^{(3,1)} = \text{diag}(1.3316, 1.1650, 1.1715, 0.8707, 1.0930, 1.0854, 0.9987, 1.2135, 1.1854, \\ 0.9195, 1.1755, 1.2645, 1.4982, 1.4077, 1.4454, 1.3744, 1.3815, 1.3340, \\ 1.3133, 1.0992, 1.0962, 1.77060)$$

For $n = 2$,

$$\hat{\Lambda}^{(3,2)} = \text{diag}(1.8566, 1.4042, 0.7619, 1.6900, 1.5335, 1.6136, 0.7826, 1.5273, \\ 1.7302, 1.8640, 1.5664, 1.6317, 1.2837, 2.0551, 1.6290)$$

For $n = 3$,

$$\hat{\Lambda}^{(3,3)} = \text{diag}(1.6139, 1.0885, 1.7570, 1.6749, 1.7878, 1.7107, \\ 1.7430, 1.7205, 2.0705, 1.7572, 2.1219, 1.7258)$$

For $n = 4$,

$$\hat{\Lambda}^{(3,4)} = \text{diag}(1.8588, 1.8938, 1.8566).$$

For $n = 5$,

$$\hat{\Lambda}^{(3,5)} = (2.0116).$$

These ordered multiplier matrices are the frozen numerical realization of the state sector in the current implementation. Their interpretation is now cleaner than before: they are no longer broad corrections on the full hadron mass. They are the realized bond multipliers of the isolated gluonic bond sector.

2.19 Final state output map

The final scalar state output is the realized product of the active multiplier channels for the relevant block. If

$$\hat{\Lambda}^{(Q,G,n)}(H) = \text{diag}(a_1, \dots, a_m), \quad \hat{\Lambda}^{(Q,G,n)} = \text{diag}(\lambda_1, \dots, \lambda_m),$$

then the realized state multiplier is

$$\mathcal{M}(H) = \prod_{j=1}^m \lambda_j^{a_j(H)}.$$

The realized bond contribution is then

$$E_{\text{state}}(H) = E_{\text{bond}}(Q) \mathcal{M}(H).$$

Substituting this into the mature mass law gives the final output structure for composite hadrons:

$$M(H) = E_{\text{bond}}(Q) \mathcal{M}(H) + \Delta E(H) + Q E_{\text{quark}}.$$

This equation is the fully formulated matrix-state version of the mature hadronic mass law.

2.20 Interpretive closure of the section

The mathematical structure developed in this section can now be read cleanly from left to right.

First, the hadron’s quark content determines its intrinsic quark baseline QE_{quark} and its cumulative transition burden $\Delta E(H)$.

Second, the hadron’s quark count selects the appropriate bond anchor:

$$E_{\text{bond}}(2) = E_{\text{base}}, \quad E_{\text{bond}}(3) = B_{\text{base}}.$$

Third, the occupancy, presence, quantum, and routing matrices determine which activation channels fire inside the state-realization machine.

Fourth, the frozen ordered multiplier matrices convert those binary activations into a realized bond multiplier $\mathcal{M}(H)$.

Finally, the hadron mass is obtained by adding the state-realized bond contribution to the transition burden and intrinsic quark contribution:

$$M(H) = E_{\text{state}}(H) + \Delta E(H) + Q E_{\text{quark}}.$$

This is the mature matrix-state architecture of the paper. Its key conceptual achievement is that it isolates the only open phenomenological layer—state realization—to the gluonic bond sector alone. The intrinsic quark baseline, the transition ladder, and the mesonic/baryonic bond anchors are now all separately grounded.

3 Dataset and Benchmark Construction

This section defines the benchmark used to evaluate the mature hadronic mass law. The purpose of the benchmark is not to optimize a narrow subset of states, but to test whether the revised decomposition

$$M(H) = E_{\text{state}}(H) + \Delta E(H) + Q E_{\text{quark}} \tag{2}$$

remains stable across a broad and heterogeneous hadronic spectrum. The dataset therefore includes light, strange, charm, and bottom hadrons, together with both mesonic and baryonic branches, so that the numerical performance of the law can be assessed against a realistic range of observed hadronic structure [1].

3.1 Benchmark scope

The benchmark is constructed to sample the main realized sectors of the observed hadron spectrum rather than a narrowly filtered subset. It includes

- single-quark baseline entries,
- mesons ($Q = 2$),
- baryons ($Q = 3$),
- light, strange, charm, and bottom flavor sectors,
- ground and excited states where descriptor assignment is sufficiently clear.

This scope is chosen so that the mass law is tested not only on the lightest anchors from which the bond scales are extracted, but also on a wide range of higher realized states.

3.2 Observed mass inputs

Observed hadron masses are taken from standard PDG reference values and associated listings [1]. These values provide the target masses against which the predictions of the present framework are compared. The benchmark therefore evaluates the internal consistency of the mature decomposition against accepted experimental hadron masses rather than against intermediate model outputs.

3.3 Descriptor structure

Each hadron in the benchmark is represented by an explicit descriptor set sufficient to evaluate all layers of the mature mass law. At minimum, the descriptor record includes

- hadron name,
- observed mass,
- quark content,
- quark count Q ,
- spin s ,
- isospin I ,
- isospin projection I_3 ,
- generation index G ,
- flavor-presence indicators Q_u, Q_d, Q_s, Q_c, Q_b ,
- heavy-flavor multiplicities N_s, N_c, N_b ,
- orbital label ℓ ,
- excitation or state label n ,
- sector-specific selectors where required.

These descriptors are not auxiliary metadata. They are the explicit inputs used by the state-realization layer.

3.4 Single-quark, mesonic, and baryonic branches

The benchmark preserves the same branch structure as the mature law itself:

$$Q = 1, \quad Q = 2, \quad Q = 3. \tag{3}$$

The $Q = 1$ entries serve as baseline quark-layer checks, since they do not involve the bond-state machinery. The $Q = 2$ and $Q = 3$ entries are the true composite sectors and therefore test the full law, including the bond anchors, the transition burden, and the matrix-state realization layer. This separation is important because it allows the quark baseline to be distinguished from the composite hadron benchmark rather than absorbed into it.

3.5 Flavor-sector coverage

The benchmark is intentionally distributed across the main flavor sectors:

- light and strange hadrons,
- charm hadrons,
- bottom hadrons.

This sector spread is necessary because the mature transition law is cumulative and the state-realization map is generation-sensitive. A benchmark restricted to only light hadrons would therefore fail to test whether the separation between intrinsic quark baseline, transition burden, and realized bond structure remains stable once heavier flavor sectors are introduced.

3.6 Excited-state coverage

Where reliable descriptor assignment is available, the benchmark includes excited hadrons in addition to ground states. This is important because the matrix-state architecture is intended to realize not only the lowest composite states but also higher orbital and excitation branches. Including excited states therefore tests whether the state-realization layer behaves as a structured map rather than as a low-state interpolation device.

3.7 Why broad coverage matters

A narrow benchmark can make a mass law appear more stable than it really is. The present benchmark is intentionally broad because the central claim of the mature framework is architectural: once intrinsic quark baseline, transition burden, and bond realization are properly separated, the same law should remain usable across a wide variety of mesonic and baryonic systems. Broad coverage is therefore part of the claim itself, not merely a presentation choice.

3.8 Prediction protocol

For each hadron H in the benchmark, the prediction protocol proceeds in the same fixed order:

1. determine the intrinsic quark baseline Q_{Equark} ,
2. compute the cumulative transition burden $\Delta E(H)$,
3. select the appropriate bond anchor through Q ,
4. evaluate the state multiplier through the descriptor-driven matrix architecture,
5. construct the predicted mass from the mature law.

This sequence mirrors the mathematical architecture developed in Section 2 and ensures that benchmark evaluation reflects the actual structure of the law rather than a collapsed shorthand implementation.

3.9 Error metric

For each hadron, the relative percent error is defined by

$$\Delta\%(H) = 100 \cdot \frac{M_{\text{pred}}(H) - M_{\text{obs}}(H)}{M_{\text{obs}}(H)}. \quad (4)$$

The principal summary statistic used throughout the paper is the mean absolute percent error over the benchmark:

$$\text{MAPE} = \frac{1}{N} \sum_{k=1}^N |\Delta\%(H_k)|. \quad (5)$$

This metric is used because it preserves the size of the relative mass deviation across the full spectrum while preventing positive and negative errors from cancelling one another.

3.10 Reproducibility and implementation transparency

The benchmark is designed to be fully reproducible from the explicit hadron list, descriptor set, constants, and matrix-state implementation. No hidden data preprocessing, implicit sector splitting, or unpublished intermediate correction layer is required to evaluate the law. This is an important part of the present framework: the benchmark is not intended as a black-box performance claim, but as a transparent test of whether the mature decomposition remains stable once implemented directly.

3.11 Role of the benchmark in the paper

The benchmark serves a precise role in the overall argument of the paper. It does not prove the conceptual interpretation by itself, but it does test whether the mature decomposition is numerically viable across a broad observed hadronic spectrum. If the revised law had improved conceptual clarity at the cost of stability, that would be immediately visible here. The benchmark therefore functions as the numerical check on the architectural claim developed in the preceding sections.

4 Results

This section reports the numerical performance of the mature hadronic mass law developed above. Because the mathematical architecture has been revised substantially, these results should be read as results of the mature formulation itself rather than as a minor update of earlier benchmarks. In particular, the state sector no longer acts on a blended prepared quantity containing both bond support and transition burden. Instead, the state layer acts only on the isolated gluonic bond scale, while the intrinsic quark baseline and cumulative flavor-transition burden remain explicit and additive. The numerical performance reported here therefore tests the revised decomposition directly.

4.1 Overall benchmark performance

Using the mature mass law

$$M(H) = Estate(H) + \Delta E(H) + Q Equark, \quad (6)$$

together with

$$Ebase = m_{\pi^0} - \Delta E_{u \rightarrow d} - 2 E_{quark}, \quad Bbase = m_p - \Delta E_{u \rightarrow d} - 3 E_{quark}, \quad a = \frac{Bbase}{Ebase}, \quad (7)$$

the present implementation achieves an average absolute percentage error of

$$\langle |\Delta\%| \rangle \approx 0.12862\%. \quad (8)$$

Only a very small number of hadrons remain above the 1% error level, and the largest residuals lie only in the upper $\sim 1.5\%$ range. The mature formulation is therefore both more accurate on average and tighter in its tail behavior than earlier published versions of the framework [6, 7].

4.2 Interpretation of the improved accuracy

The improvement matters because it is not merely numerical. It follows directly from the cleaner physical separation introduced in Section 2. Earlier versions asked the state sector to act on a quantity that already mixed together bond support and transition burden. That architecture was effective, but it also forced the state layer to absorb structure that did not properly belong to it. In the mature law, the three major contributions are now separated:

- intrinsic quark baseline,
- cumulative flavor-transition burden,
- state-realized gluonic bond energy.

The fact that the benchmark improves under this separation is strong evidence that the mature decomposition is closer to the true internal organization of the hadronic mass problem than the earlier blended form.

4.3 Mesonic sector performance

The mesonic sector is reproduced with high accuracy across light, strange, charm, and bottom families. This includes both anchor-like low states and a substantial range of higher excitations. The significance of the mesonic performance is twofold. First, it confirms that the extracted mesonic bond anchor $Ebase$ is numerically stable across a wide range of realized meson states. Second, it shows that the simplified cumulative transition ladder is sufficient even in sectors where earlier versions required more complicated rule structures.

The mesonic results are especially important because the pion anchor sits at the lowest end of the composite hierarchy. If the mature decomposition were unstable, the mesonic branch would be one of the first places where that instability would appear. Instead, the branch remains accurate across broad variation in flavor content and state label.

4.4 Baryonic sector performance

The baryonic sector is likewise reproduced with strong accuracy across light, strange, charm, and bottom baryons. This confirms that the baryonic bond anchor $Bbase$, extracted from the proton, provides a stable basis for the three-quark branch once the explicit transition burden and intrinsic quark contribution have been separated.

The baryonic benchmark is particularly important because it tests the meson–baryon scaling relation

$$a = \frac{B_{base}}{E_{base}} \quad (9)$$

under a broad range of realized three-quark configurations. The results show that the same bond architecture can support both mesonic and baryonic branches without introducing separate phenomenological mass laws for each. That is one of the central structural successes of the mature formulation.

4.5 Residual tail structure

The remaining tail cases are now much more informative than they were in earlier versions. Because the intrinsic quark baseline, transition ladder, and bond anchors are separately fixed, the dominant remaining residuals can no longer be interpreted as broad uncertainty in the whole mass law. They must instead be read mainly as limitations of the current state-realization layer.

This is an important improvement in interpretability. A residual tail is no longer evidence that the entire architecture remains unresolved. It is evidence that the open structure has been localized. In the mature law, the tail points primarily to the state matrix.

4.6 What the remaining misses mean

The remaining larger residuals should therefore not be read as failures of the foundational decomposition. They are better understood as indicators of where the present state-realization map is still too coarse. In particular, the tail suggests that some sector-specific state rules, excitation rules, or symmetry selectors are not yet fully resolved inside the matrix architecture.

That is a much stronger scientific position than a globally phenomenological model with diffuse uncertainty everywhere. The mature framework now fixes the quark baseline, the cumulative transition ladder, and the mesonic and baryonic bond anchors explicitly. What remains open is not the whole mass law, but the configuration-dependent realization of the bond sector.

4.7 Comparison with the earlier architecture

Compared with earlier published versions of the framework, the mature law improves by becoming more physically disciplined rather than more elaborate. The transition sector has been reduced to a cumulative nearest-neighbor ladder, the meson–baryon bond relation is now anchored directly to π^0 and p , and the state layer has been restricted to its proper role as the realization map of the gluonic bond sector [6, 7, 8].

This is important because it means the gain in accuracy is not coming from a larger correction apparatus. The mature formulation is more economical than the versions that preceded it. It fits better because the physical roles of the terms have been separated more cleanly.

4.8 Resulting picture of the mass problem

Taken together, the numerical results suggest a revised picture of the hadronic mass problem. The intrinsic quark baseline and flavor-transition burden appear to be comparatively rigid. The bond anchors extracted from the pion and proton also appear stable. The remaining nontrivial structure lies in how that bond sector is realized across different hadronic configurations.

In other words, the mass problem is not broadly open at every level. Most of its large-scale architecture is now fixed. The main unresolved layer is the state-dependent realization of gluonic bond structure. That is the interpretive significance of the present benchmark.

4.9 Summary of the numerical result

The mature hadronic mass law achieves sub-0.13% average absolute error across the full benchmark while simultaneously simplifying the conceptual structure of the model. It reproduces mesons and baryons within one common framework, grounded by the pion and proton anchors and organized by a cumulative transition ladder plus a state-realized bond sector. The benchmark therefore supports two conclusions at once:

1. the revised decomposition is numerically viable across a broad observed hadronic spectrum,
2. the cleaner separation of intrinsic quark baseline, transition burden, and realized bond structure is physically productive rather than merely formal.

The remaining tail behavior now points primarily to refinement of the state-realization matrix, not to a failure of the overall architecture.

5 Discussion

The mature formulation developed in this paper changes the hadronic mass problem in two important ways. First, it improves numerical performance across the benchmark. Second, and more importantly, it reorganizes the internal architecture of the problem into physically distinct layers. The intrinsic quark baseline, cumulative flavor-transition burden, and gluonic bond realization are no longer blended together. This makes the present framework easier to interpret, easier to test, and more sharply constrained than earlier versions [6, 7, 8].

5.1 What is fixed and what remains open

The benchmark results support the view that much of the large-scale mass architecture is now fixed. The intrinsic quark baseline $Q\text{ Equark}$ is explicit. The cumulative transition ladder $\Delta E(H)$ is explicit. The bond anchors $Ebase$ and $Bbase$ are extracted directly from the two lightest realized composite reference states, π^0 and p , rather than inserted as free mass constants [1, 8]. These layers therefore no longer belong to the open phenomenological sector.

What remains open is the realized state map of the bond sector. In the mature framework, the unresolved structure is not spread across the full mass law. It is concentrated in the question of how the isolated bond scale $Ebond(Q)$ is realized under the quantum, flavor, and excitation descriptors of a given hadron. This is a much narrower and more productive open problem than the earlier situation in which the state layer acted on a blended mass quantity.

5.2 Why the decomposition matters physically

The significance of the mature decomposition is therefore not only numerical. It changes the physical meaning of the terms. Intrinsic quark mass now represents what it should represent: the baseline contribution carried by constituent content. The cumulative transition ladder represents what it should represent: the explicit burden of flavor elevation. The state layer then represents only what belongs naturally to it: the realized gluonic bond structure selected by the hadron's configuration.

This separation matters because a model can fit data while still assigning the wrong physical role to its internal pieces. The present revision argues that the benchmark improves precisely because the roles of the terms have been reassigned more correctly. The state layer is no longer compensating for explicit transition structure, and the transition structure is no longer hidden inside an effective bond quantity. That is why the present form is both more economical and more interpretable than its predecessors.

5.3 Meaning of the pion and proton anchors

The use of π^0 and p as the mesonic and baryonic bond anchors is one of the central structural choices of the paper. These are not arbitrary low-mass examples; they are the two simplest realized composite reference states in the $Q = 2$ and $Q = 3$ sectors. Extracting $Ebase$ and $Bbase$ from them after subtracting intrinsic quark and transition contributions gives the bond sector a direct physical starting point.

This anchor-based construction has two advantages. First, it avoids introducing an unexplained universal mass constant. Second, it ties the bond law directly to the lowest realized composite systems rather than to an external calibration target. In this sense, the framework is built upward from physically realized hadrons rather than downward from an abstract parameter fit. The resulting ratio

$$a = \frac{Bbase}{Ebase} \quad (10)$$

then acquires a concrete interpretation as the realized meson–baryon bond scaling factor.

5.4 Interpretation of the transition ladder

Another important consequence of the mature formulation is the simplification of the transition sector. The flavor-transition burden is now represented by a cumulative nearest-neighbor ladder. This is a more economical construction than the branching or partner-dependent forms that appeared in earlier formulations [6, 7]. The present benchmark indicates that such elaboration is not required at the level of the explicit transition term itself.

This does not mean that flavor-dependent subtleties disappear. Rather, it suggests that much of the nontrivial flavor-dependent structure belongs in the state-realization layer rather than in the transition law. In the mature framework, the transition sector measures flavor elevation, while the state sector determines how the corresponding bond structure is realized for the given hadronic configuration.

5.5 What the tail cases now teach

The remaining higher-error states are now more informative than they were in earlier versions of the framework. Because the quark baseline, transition burden, and bond anchors are separately fixed, the residual discrepancies no longer point to a broad uncertainty in every part of the model. They point mainly to limitations in the present state-realization architecture.

That shift is important. Tail cases can now be read as probes of missing or overly coarse state structure: unresolved excitation rules, incomplete selector logic, sector-sensitive multiplier structure, or still-hidden symmetry patterns. In other words, the remaining error is no longer evenly distributed ignorance. It is localized information.

5.6 The status of the matrix-state architecture

The matrix-state machinery therefore occupies a very specific status in the mature framework. It is not a general-purpose correction layer appended to the whole mass law. It is the explicit realization map of the isolated bond sector. The occupancy, flavor-presence, quantum-selector, routing, activation, and ordered multiplier matrices together define how the bond anchor is transformed into the realized state contribution for each hadron.

This status is conceptually important. It means the matrix layer is not being asked to repair foundational terms that are already explicit elsewhere. Its task is narrower: to represent the structured realization of gluonic bond support. That narrowing is one of the main reasons the present formulation remains interpretable despite its broad benchmark scope.

5.7 Relation to earlier work

The present paper remains continuous with the broader program developed in the earlier published papers, but it also clarifies and sharpens that program in a decisive way. The seven-element reconstruction paper established the broader interpretive direction [6]. The variational mass and decay paper showed that a compact feature-based law could already achieve broad phenomenological reach [7]. The *Ebase* origin paper then supplied a more explicit physical account of the bond anchor itself [8]. The present work consolidates those ingredients into a cleaner final mass architecture in which the bond sector, transition burden, and intrinsic quark baseline are explicitly separated.

Because of that consolidation, the current paper should not be read simply as another incremental fit update. Its main contribution is architectural maturity. It shows that the earlier numerical success can be retained, and even improved, after the mass law has been reorganized into more physically disciplined layers.

5.8 Broader implications of the mature framework

The present results suggest that the hadronic mass problem may be more structured than many phenomenological approaches assume. Once intrinsic quark contribution, transition burden, and realized bond support are separated, the large-scale skeleton of the mass law appears comparatively rigid. That is a notable outcome. It implies that much of the remaining work may not require rethinking the whole decomposition, but refining the structured state realization of the bond sector.

This also helps clarify how the framework sits relative to the broader landscape of hadronic modeling. It does not attempt to replace QCD or its numerical formulations [3, 4, 5]. Rather, it offers a compact interpretive law that organizes observed hadron masses into explicit layers and makes the open phenomenological sector visible. In that sense, the framework is both modest and ambitious: modest in that it isolates what it does not yet solve, ambitious in that it seeks a unified structural account across a broad observed hadron spectrum.

5.9 Final perspective before extension

The main lesson of the present paper is therefore not only that the benchmark is good, but that the benchmark is good for a more intelligible reason. The mature decomposition appears to place the major physical burdens in the right locations: quark baseline in the constituent term, flavor elevation in the transition ladder, and residual configurational structure in the bond-state realization map. That is the strongest outcome of the work.

For that reason, the natural next step is not to alter the core mass law again, but to ask whether the same architectural logic can extend to higher composite systems built from already-

realized hadrons. The following section takes up that question in a deliberately limited way, not by claiming a solved exotic mass law, but by identifying the next composite layer suggested by the present framework.

6 Toward a Higher Composite Layer: Tetraquarks, Pentaquarks, and Baryon–Baryon Systems

The purpose of this section is not to extend the present results beyond their demonstrated domain, but to identify the next natural composite layer suggested by the framework developed above. The central achievement of this paper is the separation of hadronic mass into a known constituent baseline and a realized bond-state sector. That architectural separation suggests a natural question: if mesons and baryons are already realized composite hadrons, can they themselves serve as constituents of a higher composite hierarchy [1, 2, 6, 7]?

This possibility arises most directly in tetraquark, pentaquark, and baryon–baryon systems. In such cases, the relevant starting point is no longer the quark-level constituent sum used for ordinary hadrons, but the sum of already-realized hadronic composites. The problem is therefore no longer one of primary confinement, but of secondary bonding between hadronic objects whose internal structure has already been realized.

For such a system X , composed of two realized hadrons H_1 and H_2 , we define the composite mass baseline as

$$Cmass(X) = M(H_1) + M(H_2). \quad (11)$$

The observed excess or deficit relative to this composite baseline is then

$$Xbond(X) = Xmass(X) - Cmass(X). \quad (12)$$

This quantity isolates the higher composite bond sector. In direct analogy with the state-realization logic developed in the present paper, the central problem is then to determine how the inherited composite bond content is transformed into the realized exotic bond:

$$state(Cbond) = Xbond. \quad (13)$$

The full composite mass is then recovered by

$$Xmass(X) = Cmass(X) + Xbond(X). \quad (14)$$

In this picture, tetraquark and pentaquark states are not treated as fundamentally separate from the broader hadronic mass program, but as the next composite layer above mesons and baryons. A tetraquark-like system may be viewed as a meson–meson composite realization, while a pentaquark-like system may be viewed as a baryon–meson composite realization. Baryon–baryon systems belong naturally to the same hierarchy. In this sense, ordinary nuclear structure is not an exception to the composite-bond picture, but rather one of its clearest realized examples [9, 2].

The atomic mass discussion presented earlier provides an important clue here. In atomic and nuclear systems, the total mass is not simply the sum of the constituent hadrons, but the realized sum modified by a smooth collective bonding relation. At the lowest end of that curve, the simplest stable systems remain close to the constituent sum, while larger collective structures progressively depart from it. This suggests that higher hadronic composites may also follow a structured realization curve, beginning from simple pair relationships and extending toward more complex composite families [7].

A particularly useful guide is the lowest stable baryon–baryon case. A proton–neutron system provides an ideal indicator of residual strong-force overlap because the two baryons retain their internal identities to a very high degree while their long-range strong tails overlap to form a stable higher composite. In such an ideal condition, the inherited composite bond is carried forward with little distortion. This motivates the working hypothesis that, for the simplest and most balanced higher composites, the state map may act approximately as an identity:

$$Xbond \approx Cbond. \quad (15)$$

Departures from this ideal relation would then encode the higher composite state rules.

Although the data remain incomplete, two state ingredients appear particularly compelling at this stage: spin and electric charge. Spin is already known throughout atomic and hadronic structure to govern whether composite configurations are balanced or frustrated. Charge is equally important at larger composite distances, where electromagnetic effects, though weaker than primary confinement, are no longer negligible relative to the small residual bond sector being isolated here. In this sense, charge need not dominate the higher composite bond; it need only be large enough to perturb it. A balanced state may therefore be one in which the inherited bond is minimally disturbed, whereas spin imbalance, charge asymmetry, higher charge magnitude, or unfavorable pairing conditions may shift the realized system away from the ideal carry-through relation [7, 1].

This interpretation naturally suggests that currently observed tetraquark and pentaquark states may represent only the lowest visible members of a broader composite hierarchy. From this perspective, exotic hadrons are not isolated anomalies, but the first experimentally accessible examples of a more general hadronic bonding structure. If future colliders reach sufficient energies and production densities not only to create exotic hadrons but also to allow them to interact, bond, and scale into larger systems, then an entire spectrum of atomic-like exotic composite structures may await discovery.

The present paper does not claim a solved mass law for such systems. Rather, it establishes the architectural foundation from which that next layer can be approached: first isolate the known composite baseline, then isolate the residual bond sector, and finally determine the state rules that govern the realized higher composite mass. In this way, tetraquarks, pentaquarks, and baryon–baryon systems arise not as a departure from the present framework, but as its next natural continuation [6, 7, 8].

7 Conclusion

This paper has presented a mature hadronic mass law in which intrinsic quark baseline, cumulative flavor-transition burden, and gluonic bond realization are separated into distinct physical layers. The resulting formulation,

$$M(H) = Estate(H) + \Delta E(H) + Q Equark, \quad (16)$$

assigns each contribution a clear role: the intrinsic quark term provides the constituent baseline, the transition ladder accounts for explicit flavor elevation, and the state sector realizes the gluonic bond structure selected by the hadron’s configuration.

Within this architecture, the mesonic and baryonic bond anchors are extracted directly from the neutral pion and proton, and their ratio defines a unified meson–baryon bond scaling law. The cumulative transition ladder provides a compact and explicit representation of flavor elevation, while the matrix-state framework realizes the bond sector through structured activation and multiplier maps.

The benchmark results show that this decomposition is not only conceptually cleaner, but numerically stable across a broad hadronic spectrum. The intrinsic quark baseline, transition burden, and bond anchors appear to be well constrained within the present framework. The primary remaining freedom is localized to the state-realization layer of the gluonic bond sector.

This localization is the central outcome of the work. It indicates that the hadronic mass problem is not uniformly open, but structured: much of its large-scale architecture is fixed, while the remaining refinement lies in the detailed realization of bond structure across different quantum configurations.

In this sense, the present formulation should be understood as a completed structural foundation rather than a final closed model. It establishes a framework in which further progress can be directed specifically at the state-realization map, and in which higher composite extensions, including tetraquark, pentaquark, and baryon–baryon systems, arise naturally as the next domain of investigation.

A Python Source Code with 6 quarks and 275 hadrons dataset

```
"""
Developed by Thomas Lock
Hadronic Mass from Confinement, Transition Structure, Matrix State Realization
"""

from dataclasses import dataclass
from math import pow

@dataclass
class Hadron:
    name: str
    mass: float
    spin: float
    isospin: float
    isospin3: float
    quarks: list[str]
    state: int
    sdsymmetry: float = 0.0
    orbital: int = 0 # S = 0, P = 1, D = 2

hadrons: list[Hadron] = [
    # Quarks
    Hadron(name="u", mass=2.1700, spin=0.5, isospin=0.5, isospin3=0.5, quarks=["u"], state=1),
    Hadron(name="d", mass=4.6700, spin=0.5, isospin=0.5, isospin3=-0.5, quarks=["d"], state=1),
    Hadron(name="s", mass=93.4000, spin=0.5, isospin=0.0, isospin3=0.0, quarks=["s"], state=1),
    Hadron(name="c", mass=1271.0000, spin=0.5, isospin=0.0, isospin3=0.0, quarks=["c"], state=1),
    Hadron(name="b", mass=4180.0000, spin=0.5, isospin=0.0, isospin3=0.0, quarks=["b"], state=1),
    Hadron(name="t", mass=172500.00000, spin=0.5, isospin=0.0, isospin3=0.0, quarks=["t"], state=1),

    # Mesons
    Hadron(name="Pi0", mass=134.9766, spin=0.0, isospin=1.0, isospin3=0.0, quarks=["u", "d"], state=1),
    Hadron(name="Pi+", mass=139.57039, spin=0.0, isospin=1.0, isospin3=1.0, quarks=["u", "d"], state=1),
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    Hadron(name="K0_L", mass=497.611, spin=0.0, isospin=0.5, isospin3=0.5, quarks=["d", "s"], state=1),
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    Hadron(name="Rho+", mass=775.11, spin=1.0, isospin=1.0, isospin3=1.0, quarks=["u", "d"], state=1),
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 Hadron(name="BJ(6114)0", mass=6114.0, spin=2.0, isospin=0.0, isospin3=0.0, quarks=["b", "s"], state=4),
 Hadron(name="BsJ(6114)0bar", mass=6114.0, spin=2.0, isospin=0.0, isospin3=0.0, quarks=["b", "s"], state=4),

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Hadron(name="Bc+", mass=6274.9, spin=0.0, isospin=0.0, isospin3=0.0, quarks=["b", "c"], state=1),
Hadron(name="Bc-", mass=6274.9, spin=0.0, isospin=0.0, isospin3=0.0, quarks=["b", "c"], state=1),
Hadron(name="Bc(2S)*+", mass=6875.0, spin=1.0, isospin=0.0, isospin3=0.0, quarks=["b", "c"], state=2),
Hadron(name="Bc(2S)*-", mass=6875.0, spin=1.0, isospin=0.0, isospin3=0.0, quarks=["b", "c"], state=2),
Hadron(name="Eta_b(1S)", mass=9399.0, spin=0.0, isospin=0.0, isospin3=0.0, quarks=["b", "b"], state=1),
Hadron(name="Upsilon(1S)", mass=9460.3, spin=1.0, isospin=0.0, isospin3=0.0, quarks=["b", "b"], state=1),
Hadron(name="Chi_b0(1P)", mass=9859.4, spin=0.0, isospin=0.0, isospin3=0.0, quarks=["b", "b"], state=1, orbital=1),
Hadron(name="Chi_b1(1P)", mass=9892.8, spin=1.0, isospin=0.0, isospin3=0.0, quarks=["b", "b"], state=1, orbital=1),
Hadron(name="h_b(1P)", mass=9899.3, spin=1.0, isospin=0.0, isospin3=0.0, quarks=["b", "b"], state=1, orbital=1),
Hadron(name="Chi_b2(1P)", mass=9912.2, spin=2.0, isospin=0.0, isospin3=0.0, quarks=["b", "b"], state=1, orbital=1),
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Hadron(name="Upsilon(2S)", mass=10023.26, spin=1.0, isospin=0.0, isospin3=0.0, quarks=["b", "b"], state=2),
Hadron(name="Chi_b0(2P)", mass=10232.5, spin=0.0, isospin=0.0, isospin3=0.0, quarks=["b", "b"], state=2, orbital=1),
Hadron(name="Chi_b1(2P)", mass=10255.5, spin=1.0, isospin=0.0, isospin3=0.0, quarks=["b", "b"], state=2, orbital=1),
Hadron(name="h_b(2P)", mass=10259.8, spin=1.0, isospin=0.0, isospin3=0.0, quarks=["b", "b"], state=2, orbital=1),
Hadron(name="Chi_b2(2P)", mass=10268.7, spin=2.0, isospin=0.0, isospin3=0.0, quarks=["b", "b"], state=2, orbital=1),
Hadron(name="Upsilon(3S)", mass=10355.2, spin=1.0, isospin=0.0, isospin3=0.0, quarks=["b", "b"], state=3),
Hadron(name="Chi_b0(3P)", mass=10495.0, spin=0.0, isospin=0.0, isospin3=0.0, quarks=["b", "b"], state=3, orbital=1),
Hadron(name="Chi_b1(3P)", mass=10513.4, spin=1.0, isospin=0.0, isospin3=0.0, quarks=["b", "b"], state=3, orbital=1),
Hadron(name="Chi_b2(3P)", mass=10524.0, spin=2.0, isospin=0.0, isospin3=0.0, quarks=["b", "b"], state=3, orbital=1),
Hadron(name="Upsilon(4S)", mass=10579.4, spin=1.0, isospin=0.0, isospin3=0.0, quarks=["b", "b"], state=4),
Hadron(name="Upsilon(5S)", mass=10885.0, spin=1.0, isospin=0.0, isospin3=0.0, quarks=["b", "b"], state=5),
Hadron(name="Upsilon(6S)", mass=11000.0, spin=1.0, isospin=0.0, isospin3=0.0, quarks=["b", "b"], state=6),

# Baryons
Hadron(name="Proton", mass=938.272089, spin=0.5, isospin=0.5, isospin3=0.5, quarks=["u", "u", "d"], state=1),
Hadron(name="Neutron", mass=939.565421, spin=0.5, isospin=0.5, isospin3=-0.5, quarks=["u", "d", "d"], state=1),
Hadron(name="Lambda0", mass=1115.683, spin=0.5, isospin=0.0, isospin3=0.0, quarks=["u", "d", "s"], state=1),
Hadron(name="Sigma+", mass=1189.370, spin=0.5, isospin=1.0, isospin3=1.0, quarks=["u", "u", "s"], state=1),
Hadron(name="Sigma0", mass=1192.642, spin=0.5, isospin=1.0, isospin3=0.0, quarks=["u", "d", "s"], state=1),
Hadron(name="Sigma-", mass=1197.449, spin=0.5, isospin=1.0, isospin3=-1.0, quarks=["d", "d", "s"], state=1),
Hadron(name="Delta++", mass=1232.000, spin=1.5, isospin=1.5, isospin3=1.5, quarks=["u", "u", "u"], state=1),
Hadron(name="Delta+", mass=1232.000, spin=1.5, isospin=1.5, isospin3=0.5, quarks=["u", "u", "d"], state=1),
Hadron(name="Delta0", mass=1232.000, spin=1.5, isospin=1.5, isospin3=-0.5, quarks=["u", "d", "d"], state=1),
Hadron(name="Delta-", mass=1232.000, spin=1.5, isospin=1.5, isospin3=-1.5, quarks=["d", "d", "d"], state=1),
Hadron(name="Xi0", mass=1314.860, spin=0.5, isospin=0.5, isospin3=0.5, quarks=["u", "s", "s"], state=1),
Hadron(name="Xi-", mass=1321.710, spin=0.5, isospin=0.5, isospin3=-0.5, quarks=["d", "s", "s"], state=1),
Hadron(name="Sigma*+", mass=1382.8, spin=1.5, isospin=1.0, isospin3=1.0, quarks=["u", "u", "s"], state=1),
Hadron(name="Sigma*0", mass=1383.7, spin=1.5, isospin=1.0, isospin3=0.0, quarks=["u", "d", "s"], state=1),
Hadron(name="Sigma*-", mass=1387.2, spin=1.5, isospin=1.0, isospin3=-1.0, quarks=["d", "d", "s"], state=1),
Hadron(name="Lambda(1405)0", mass=1405.1, spin=0.5, isospin=0.0, isospin3=0.0, quarks=["u", "d", "s"], state=2),
Hadron(name="Lambda(1520)0", mass=1519.5, spin=1.5, isospin=0.0, isospin3=0.0, quarks=["u", "d", "s"], state=2),
Hadron(name="Xi*0", mass=1531.80, spin=1.5, isospin=0.5, isospin3=0.5, quarks=["u", "s", "s"], state=1),
Hadron(name="Xi*-", mass=1535.0, spin=1.5, isospin=0.5, isospin3=-0.5, quarks=["d", "s", "s"], state=1),
Hadron(name="Lambda(1600)0", mass=1600.0, spin=0.5, isospin=0.0, isospin3=0.0, quarks=["u", "d", "s"], state=3),
Hadron(name="Omega", mass=1672.450, spin=1.5, isospin=0.0, isospin3=0.0, quarks=["s", "s", "s"], state=1),
Hadron(name="Lambda(1690)0", mass=1690.0, spin=1.5, isospin=0.0, isospin3=0.0, quarks=["u", "d", "s"], state=3),
Hadron(name="Lambda(1820)0", mass=1820.0, spin=2.5, isospin=0.0, isospin3=0.0, quarks=["u", "d", "s"], state=4),
Hadron(name="Xi(1690)0", mass=1690.0, spin=0.5, isospin=0.5, isospin3=0.5, quarks=["u", "s", "s"], state=2),
Hadron(name="Xi(1690)-", mass=1690.0, spin=0.5, isospin=0.5, isospin3=-0.5, quarks=["d", "s", "s"], state=2),
Hadron(name="Xi(1820)0", mass=1823.0, spin=1.5, isospin=0.5, isospin3=0.5, quarks=["u", "s", "s"], state=3),
Hadron(name="Xi(1820)-", mass=1823.0, spin=1.5, isospin=0.5, isospin3=-0.5, quarks=["d", "s", "s"], state=3),
Hadron(name="Xi(1950)0", mass=1950.0, spin=0.5, isospin=0.5, isospin3=0.5, quarks=["u", "s", "s"], state=4),
Hadron(name="Xi(1950)-", mass=1950.0, spin=0.5, isospin=0.5, isospin3=-0.5, quarks=["d", "s", "s"], state=4),
Hadron(name="Omega(2012)-", mass=2012.4, spin=1.5, isospin=0.0, isospin3=0.0, quarks=["s", "s", "s"], state=2),
Hadron(name="Omega(2250)-", mass=2252.0, spin=2.5, isospin=0.0, isospin3=0.0, quarks=["s", "s", "s"], state=3),
Hadron(name="Lambda+c", mass=2286.460, spin=0.5, isospin=0.0, isospin3=0.0, quarks=["u", "d", "c"], state=1),
Hadron(name="Sigma+c", mass=2453.970, spin=0.5, isospin=1.0, isospin3=1.0, quarks=["u", "u", "c"], state=1),
Hadron(name="Sigma+c", mass=2452.900, spin=0.5, isospin=1.0, isospin3=0.0, quarks=["u", "d", "c"], state=1),
Hadron(name="Sigma0c", mass=2453.750, spin=0.5, isospin=1.0, isospin3=-1.0, quarks=["d", "d", "c"], state=1),
Hadron(name="Xi0c", mass=2470.850, spin=0.5, isospin=0.5, isospin3=-0.5, quarks=["d", "s", "c"], state=1),
Hadron(name="Xi+c", mass=2468.000, spin=0.5, isospin=0.5, isospin3=0.5, quarks=["u", "s", "c"], state=1),
Hadron(name="Sigma*++c", mass=2518.4, spin=1.5, isospin=1.0, isospin3=1.0, quarks=["u", "u", "c"], state=1),
Hadron(name="Sigma*+c", mass=2517.5, spin=1.5, isospin=1.0, isospin3=0.0, quarks=["u", "d", "c"], state=1),
Hadron(name="Sigma*0c", mass=2518.1, spin=1.5, isospin=1.0, isospin3=-1.0, quarks=["d", "d", "c"], state=1),
Hadron(name="Xi'+c", mass=2578.4, spin=0.5, isospin=0.5, isospin3=0.5, quarks=["u", "s", "c"], state=1, sdsymmetry=1.0),
Hadron(name="Xi'0c", mass=2578.8, spin=0.5, isospin=0.5, isospin3=-0.5, quarks=["d", "s", "c"], state=1, sdsymmetry=1.0),
Hadron(name="Lambda(2595)+c", mass=2592.25, spin=0.5, isospin=0.0, isospin3=0.0, quarks=["u", "d", "c"], state=2),

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Hadron(name="Lambda(2625)+c", mass=2628.11, spin=1.5, isospin=0.0, isospin3=0.0, quarks=["u", "d", "c"], state=2),
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Hadron(name="Xi*(2645)0c", mass=2646.3, spin=1.5, isospin=0.5, isospin3=-0.5, quarks=["d", "s", "c"], state=1),
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Hadron(name="Omega*c", mass=2765.9, spin=1.5, isospin=0.0, isospin3=0.0, quarks=["s", "s", "c"], state=1),
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Hadron(name="Xi(2790)0c", mass=2793.9, spin=0.5, isospin=0.5, isospin3=-0.5, quarks=["d", "s", "c"], state=2),
Hadron(name="Sigma(2800)+c", mass=2792.0, spin=1.5, isospin=1.0, isospin3=0.0, quarks=["u", "d", "c"], state=2),
Hadron(name="Sigma(2800)++c", mass=2801.0, spin=1.5, isospin=1.0, isospin3=1.0, quarks=["u", "u", "c"], state=2),
Hadron(name="Sigma(2800)0c", mass=2806.0, spin=1.5, isospin=1.0, isospin3=-1.0, quarks=["d", "d", "c"], state=2),
Hadron(name="Xi(2815)+c", mass=2816.6, spin=1.5, isospin=0.5, isospin3=0.5, quarks=["u", "s", "c"], state=2),
Hadron(name="Xi(2815)0c", mass=2819.85, spin=1.5, isospin=0.5, isospin3=-0.5, quarks=["d", "s", "c"], state=2),
Hadron(name="Lambda(2860)+c", mass=2856.1, spin=1.5, isospin=0.0, isospin3=0.0, quarks=["u", "d", "c"], state=3),
Hadron(name="Lambda(2880)+c", mass=2881.6, spin=2.5, isospin=0.0, isospin3=0.0, quarks=["u", "d", "c"], state=3),
Hadron(name="Xi(2882)+c", mass=2882.0, spin=2.5, isospin=0.5, isospin3=0.5, quarks=["u", "s", "c"], state=2),
Hadron(name="Xi(2882)0c", mass=2882.0, spin=2.5, isospin=0.5, isospin3=-0.5, quarks=["d", "s", "c"], state=2),
Hadron(name="Lambda(2910)+c", mass=2910.6, spin=0.5, isospin=0.0, isospin3=0.0, quarks=["u", "d", "c"], state=3),
Hadron(name="Lambda(2940)+c", mass=2939.6, spin=1.5, isospin=0.0, isospin3=0.0, quarks=["u", "d", "c"], state=3),
Hadron(name="Xi(2923)+c", mass=2923.0, spin=0.5, isospin=0.5, isospin3=0.5, quarks=["u", "s", "c"], state=3),
Hadron(name="Xi(2923)0c", mass=2923.0, spin=0.5, isospin=0.5, isospin3=-0.5, quarks=["d", "s", "c"], state=3),
Hadron(name="Xi(2930)+c", mass=2930.0, spin=0.5, isospin=0.5, isospin3=0.5, quarks=["u", "s", "c"], state=3),
Hadron(name="Xi(2930)0c", mass=2930.0, spin=0.5, isospin=0.5, isospin3=-0.5, quarks=["d", "s", "c"], state=3),
Hadron(name="Xi(2970)+c", mass=2967.0, spin=1.5, isospin=0.5, isospin3=0.5, quarks=["u", "s", "c"], state=3),
Hadron(name="Xi(2970)0c", mass=2967.0, spin=1.5, isospin=0.5, isospin3=-0.5, quarks=["d", "s", "c"], state=3),
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Hadron(name="Xi(3080)+c", mass=3077.2, spin=2.5, isospin=0.5, isospin3=0.5, quarks=["u", "s", "c"], state=4),
Hadron(name="Omega(3000)0c", mass=3000.4, spin=0.5, isospin=0.0, isospin3=0.0, quarks=["s", "s", "c"], state=2),
Hadron(name="Omega(3050)0c", mass=3050.2, spin=0.5, isospin=0.0, isospin3=0.0, quarks=["s", "s", "c"], state=2),
Hadron(name="Omega(3065)0c", mass=3065.6, spin=1.5, isospin=0.0, isospin3=0.0, quarks=["s", "s", "c"], state=2),
Hadron(name="Omega(3090)0c", mass=3090.2, spin=0.5, isospin=0.0, isospin3=0.0, quarks=["s", "s", "c"], state=3),
Hadron(name="Omega(3120)0c", mass=3119.1, spin=1.5, isospin=0.0, isospin3=0.0, quarks=["s", "s", "c"], state=3),
Hadron(name="Xi(3123)+c", mass=3122.9, spin=2.5, isospin=0.5, isospin3=0.5, quarks=["u", "s", "c"], state=4),
Hadron(name="Xi(3123)0c", mass=3122.9, spin=2.5, isospin=0.5, isospin3=-0.5, quarks=["d", "s", "c"], state=4),
Hadron(name="Omega(3185)0c", mass=3185.1, spin=0.5, isospin=0.0, isospin3=0.0, quarks=["s", "s", "c"], state=4),
Hadron(name="Omega(3327)0c", mass=3327.1, spin=1.5, isospin=0.0, isospin3=0.0, quarks=["s", "s", "c"], state=5),
Hadron(name="Sigma++cc", mass=3621.7, spin=0.5, isospin=0.5, isospin3=-0.5, quarks=["u", "c", "c"], state=1),
Hadron(name="Sigma+cc", mass=3619.97, spin=0.5, isospin=0.5, isospin3=0.5, quarks=["d", "c", "c"], state=1),
Hadron(name="Omega+cc", mass=3730.97, spin=0.5, isospin=0.0, isospin3=0.0, quarks=["s", "c", "c"], state=1),
Hadron(name="Lambda0b", mass=5619.600, spin=0.5, isospin=0.0, isospin3=0.0, quarks=["u", "d", "b"], state=1),
Hadron(name="Xi0b", mass=5787.800, spin=0.5, isospin=0.5, isospin3=0.5, quarks=["u", "s", "b"], state=1),
Hadron(name="Xi1b", mass=5794.400, spin=0.5, isospin=0.5, isospin3=-0.5, quarks=["d", "s", "b"], state=1),
Hadron(name="Sigma+b", mass=5810.55, spin=0.5, isospin=1.0, isospin3=1.0, quarks=["u", "u", "b"], state=1),
Hadron(name="Sigma0b", mass=5813.4, spin=0.5, isospin=1.0, isospin3=0.0, quarks=["u", "d", "b"], state=1),
Hadron(name="Sigma-b", mass=5815.5, spin=0.5, isospin=1.0, isospin3=-1.0, quarks=["d", "d", "b"], state=1),
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Hadron(name="Sigma*0b", mass=5831.8, spin=1.5, isospin=1.0, isospin3=0.0, quarks=["u", "d", "b"], state=1),
Hadron(name="Sigma*-b", mass=5834.7, spin=1.5, isospin=1.0, isospin3=-1.0, quarks=["d", "d", "b"], state=1),
Hadron(name="Lambda(5912)0b", mass=5912.2, spin=0.5, isospin=0.0, isospin3=0.0, quarks=["u", "d", "b"], state=2),
Hadron(name="Lambda(5920)0b", mass=5920.1, spin=1.5, isospin=0.0, isospin3=0.0, quarks=["u", "d", "b"], state=2),
Hadron(name="Xi'0b", mass=5935.9, spin=0.5, isospin=0.5, isospin3=0.5, quarks=["u", "s", "b"], state=1, sdsymmetry=1.0),
Hadron(name="Xi'-b", mass=5935.0, spin=0.5, isospin=0.5, isospin3=-0.5, quarks=["d", "s", "b"], state=1, sdsymmetry=1.0),
Hadron(name="Xi(5945)0b", mass=5945.3, spin=1.5, isospin=0.5, isospin3=0.5, quarks=["u", "s", "b"], state=1),
Hadron(name="Xi*(5955)-b", mass=5955.3, spin=1.5, isospin=0.5, isospin3=-0.5, quarks=["d", "s", "b"], state=1),
Hadron(name="Omegab", mass=6045.100, spin=0.5, isospin=0.0, isospin3=0.0, quarks=["s", "s", "b"], state=1),
Hadron(name="Omega*b", mass=6045.2, spin=1.5, isospin=0.0, isospin3=0.0, quarks=["s", "s", "b"], state=1),
Hadron(name="Lambda(6070)0b", mass=6072.0, spin=0.5, isospin=0.0, isospin3=0.0, quarks=["u", "d", "b"], state=3),
Hadron(name="Sigma(6097)+b", mass=6095.8, spin=1.5, isospin=1.0, isospin3=1.0, quarks=["u", "u", "b"], state=2),
Hadron(name="Sigma(6097)-b", mass=6098.0, spin=1.5, isospin=1.0, isospin3=-1.0, quarks=["d", "d", "b"], state=2),
Hadron(name="Lambda(6146)0b", mass=6146.2, spin=1.5, isospin=0.0, isospin3=0.0, quarks=["u", "d", "b"], state=3),
Hadron(name="Lambda(6152)0b", mass=6152.5, spin=2.5, isospin=0.0, isospin3=0.0, quarks=["u", "d", "b"], state=3),

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]

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transitionL1: dict[str, float] = {
    "u-d": 2.50,
    "d-s": 88.73,
    "s-c": 1177.60,
    "c-b": 2909.00,

```

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    "b+t": 168320.00,
}

generation: dict[str, float] = {
    "u": 1, "d": 1,
    "s": 2, "c": 2,
    "b": 3, "t": 3
}

E_quark = 2.17000
pion_mass = 134.9766
B_base = 938.272089 - transitionL1["u+d"] - 3 * E_quark
E_base = pion_mass - transitionL1["u+d"] - 2 * E_quark

def deltaEbase(quarks: list[str]) -> float:
    total = 0.0
    for q in quarks:
        if q == "u":
            total += 0
        elif q == "d":
            total += transitionL1["u+d"]
        elif q == "s":
            total += transitionL1["u+d"] + transitionL1["d+s"]
        elif q == "c":
            total += transitionL1["u+d"] + transitionL1["d+s"] + transitionL1["s+c"]
        elif q == "b":
            total += transitionL1["u+d"] + transitionL1["d+s"] + transitionL1["s+c"] + transitionL1["c+b"]
        elif q == "t":
            total += transitionL1["u+d"] + transitionL1["d+s"] + transitionL1["s+c"] + transitionL1["c+b"] + transitionL1["b+t"]
        else:
            total += 0
    return total

def gate(condition: bool) -> float:
    return 1.0 if condition else 0.0

def stateEnergy(E_debase: float, state: float, spin: float, isospin: float, isospin3: float,
    sdsymmetry: float, G: float, Q: float, Qu: float, Qd: float, Qs: float,
    Qc: float, Qb: float, Ns: float, Nc: float, Nb: float, orbital: int) -> float:
    stateExcitation = 0.0

    if Q == 2:
        if G == 1:
            if state == 0:
                stateExcitation = E_debase * 7.5947
            elif state == 1:
                stateExcitation = (
                    E_debase
                    * pow(4.2222, gate(spin == 0 and isospin == 0 and abs(isospin3) == 0))
                    * pow(1.0358, gate(spin == 0 and isospin == 1 and abs(isospin3) == 1))
                    * pow(5.9957, gate(spin == 1))
                    * pow(10.2340, gate(spin == 2))
                )
            elif state == 2:
                stateExcitation = (
                    E_debase
                    * pow(11.4097, gate(isospin == 1))
                    * pow(10.0452, gate(spin == 0 and isospin == 0))
                    * pow(10.9505, gate(spin == 1 and isospin == 0))
                    * pow(9.9008, gate(spin == 2 and isospin == 0))
                )
            elif state == 3:
                stateExcitation = (
                    E_debase
                    * pow(13.6117, gate(spin == 0 and isospin == 0))

```



```

        * pow(14.0878, gate(spin == 0 and isospin == 1))
        * pow(12.9795, gate(spin == 1 and isospin == 0))
        * pow(13.3697, gate(spin == 1 and isospin == 1))
        * pow(13.1263, gate(spin == 3))
    )
elif G == 2:
    if state == 1:
        stateExcitation = (
            E_debase
            * pow(1.3037, gate(spin == 0.0 and isospin == 0.0) * Qs)
            * pow(3.6152, gate(spin == 0.0 and isospin == 0.0) * Qs * Qc)
            * pow(3.1180, gate(spin == 0.0 and isospin == 0.5) * Qs)
            * pow(4.6275, gate(spin == 0.0 and isospin == 0.5) * Qc)
            * pow(5.8359, gate(spin == 1.0 and isospin == 0.0) * Qc * Qs)
            * pow(6.4982, gate(spin == 1.0 and isospin == 0.0 and Ns == 2))
            * pow(6.2129, gate(spin == 1.0 and isospin == 0.5) * Qs)
            * pow(5.7257, gate(spin == 1.0 and isospin == 0.5) * Qc)
            * pow(4.6152, gate(spin == 0.0 and isospin == 0 and Ns == 2))
            * pow(3.4486, gate(spin == 0.0 and orbital == 0 and Nc == 2))
            * pow(4.3305, gate(spin == 1.0 and orbital == 0 and Nc == 2))
            * pow(6.8108, gate(spin == 0.0 and orbital == 1 and Nc == 2))
            * pow(7.6744, gate(spin == 1.0 and orbital == 1 and Nc == 2))
            * pow(7.9147, gate(spin == 2.0 and orbital == 1 and Nc == 2))
        )
    elif state == 2:
        stateExcitation = (
            E_debase
            * pow(7.4404, gate(spin == 0 and isospin == 0.0))
            * pow(8.1540, gate(spin == 0 and isospin == 0.5) * Qu)
            * pow(8.7978, gate(spin == 0 and isospin == 0.5) * Qd)
            * pow(8.8363, gate(spin == 1 and isospin == 0.0) * Qc * Qs)
            * pow(11.653, gate(spin == 1 and isospin == 0.0 and Ns == 2))
            * pow(9.42899, gate(spin == 2 and isospin == 0.0) * Qc * Qs)
            * pow(8.9770, gate(spin == 1 and isospin == 0.5))
            * pow(9.2825, gate(spin == 2 and isospin == 0.5))
            * pow(1.1490, gate(spin == 0 and Nc == 2))
            * pow(8.9287, gate(spin == 1 and Nc == 2))
        )
    elif state == 3:
        stateExcitation = (
            E_debase
            * pow(13.0111, gate(Ns == 2))
            * pow(9.5725, gate(spin == 0 and isospin == 0.0))
            * pow(10.5638, gate(spin == 1 and isospin == 0.5))
            * pow(10.57332, gate(spin == 1 and isospin == 0.0) * Qc * Qs)
            * pow(9.6467, gate(spin == 1 and isospin == 0.0 and Nc == 2))
            * pow(9.9909, gate(spin == 2 and isospin == 0.0 and Nc == 2))
            * pow(10.1508, gate(spin == 3 and isospin == 0.0 and Nc == 2))
        )
    elif state == 4:
        stateExcitation = (
            E_debase
            * pow(15.8752, gate(Ns == 2))
            * pow(11.6308, Qc * Qu)
            * pow(11.6691, Qc * Qs)
            * pow(11.6828, gate(Nc == 2))
        )
    elif state == 5:
        stateExcitation = (
            E_debase
            * pow(13.1078, Qs * Qc)
            * pow(12.8690, gate(Nc == 2))
        )
    elif state == 6:
        stateExcitation = E_debase * 14.6640
elif G == 3:
    if state == 1:
        stateExcitation = (

```

```

        E_debase
        * pow(6.4298, gate(spin == 0 and isospin == 0.0) * Qc)
        * pow(8.5337, gate(spin == 0 and isospin == 0.0) * Qs)
        * pow(8.1085, gate(spin == 0 and isospin == 0.0 and Nb == 2 and orbital == 0))
        * pow(8.5606, gate(spin == 0 and isospin == 0.5))
        * pow(8.9123, gate(spin == 1 and isospin == 0.0) * Qs)
        * pow(8.5869, gate(spin == 1 and isospin == 0.0 and Nb == 2 and orbital == 0))
        * pow(8.9165, gate(spin == 1 and isospin == 0.5))
        * pow(11.9622, gate(Nb == 2 and orbital == 1))
    )
elif state == 2:
    stateExcitation = (
        E_debase
        * pow(12.7910, gate(spin == 0 and isospin == 0.0 and Nb == 2 and orbital == 0))
        * pow(14.6133, gate(spin == 0 and isospin == 0.0 and Nb == 2 and orbital == 1))
        * pow(12.9803, gate(spin == 1 and isospin == 0.0 and Nb == 2 and orbital == 0))
        * pow(14.826, gate(spin == 1 and isospin == 0.0 and Nb == 2 and orbital == 1))
        * pow(14.8958, gate(spin == 2 and isospin == 0.0 and Nb == 2))
        * pow(12.0388, Qu + Qd)
        * pow(12.1798, Qs)
        * pow(11.1131, Qc)
    )
elif state == 3:
    stateExcitation = (
        E_debase
        * pow(16.0882, gate(Nb == 2))
        * pow(13.0958, gate(spin == 1.0 and isospin == 0.5))
        * pow(12.3040, gate(spin == 1 and isospin == 0) * Qs * Qb)
    )
elif state == 4:
    stateExcitation = (
        E_debase
        * pow(13.8823, Qu)
        * pow(13.9408, Qd)
        * pow(13.9663, Qs)
        * pow(1.0285, gate(spin == 2) * Qs)
        * pow(17.3206, gate(Nb == 2))
    )
elif state == 5:
    stateExcitation = E_debase * 19.7055
elif state == 6:
    stateExcitation = E_debase * 20.6030
elif Q == 3:
    if state == 1:
        stateExcitation = (
            E_debase
            * pow(1.3316, gate(spin == 0.5 and Ns == 2) * Qc)
            * pow(1.1650, gate(spin == 0.5 and Nc == 2))
            * pow(1.1715, gate(spin == 0.5 and isospin == 0.0 and Ns == 2) * Qb)
            * pow(0.8707, gate(spin == 0.5 and isospin == 0.0) * Qb)
            * pow(1.093, gate(spin == 0.5 and isospin == 0.0) * Qd * Qs)
            * pow(1.0854, gate(spin == 0.5 and isospin == 0.0) * Qd * Qc)
            * pow(0.9987, gate(spin == 0.5 and isospin == 0.5 and isospin3 == -0.5) * Qu * Qd)
            * pow(1.2135, gate(spin == 0.5 and isospin == 0.5 and Ns == 2))
            * pow(1.1854, gate(spin == 0.5 and isospin == 0.5) * Qs * Qc)
            * pow(0.9195, gate(spin == 0.5 and isospin == 0.5) * Qb)
            * pow(1.1755, gate(spin == 0.5 and isospin == 1.0) * Qs)
            * pow(1.2645, gate(spin == 0.5 and isospin == 1.0) * Qc)
            * pow(1.4982, gate(spin == 1.5 and isospin == 0.0 and Ns == 3))
            * pow(1.4077, gate(spin == 1.5 and isospin == 0.0) * Qc)
            * pow(1.4454, gate(spin == 1.5 and isospin == 0.5 and Ns == 2))
            * pow(1.3744, gate(spin == 1.5 and isospin == 0.5) * Qc)
            * pow(1.3815, gate(spin == 1.5 and isospin == 1.0) * Qs)
            * pow(1.3340, gate(spin == 1.5 and isospin == 1.0) * Qc)
            * pow(1.3133, gate(spin == 1.5 and isospin == 1.5))
            * pow(1.0992, gate(sdsymmetry == 1.0) * Qc)
            * pow(1.0962, gate(sdsymmetry == 1.0) * Qb)
            * pow(1.77060, Qb)
        )

```

```

    )
elif state == 2:
    stateExcitation = (
        E_debase
        * pow(1.8566, gate(isospin == 0.0 and abs(isospin3) == 0.0 and Ns == 0) * Qd * Qu)
        * pow(1.4042, gate(spin == 0.5 and isospin == 0.0 and abs(isospin3) == 0.0) * Qs * Qd * Qu)
        * pow(0.7619, gate(spin == 0.5 and isospin == 0.0 and abs(isospin3) == 0.0) * Qc * Qd * Qu)
        * pow(1.6900, gate(spin == 0.5 and isospin == 0.0 and Ns == 2) * Qc)
        * pow(1.5335, gate(spin == 0.5 and isospin == 0.5 and Ns == 1))
        * pow(1.6136, gate(spin == 0.5 and isospin == 0.5 and Ns == 2))
        * pow(0.7826, gate(spin == 1.5 and isospin == 0.0 and Ns == 0))
        * pow(1.5273, gate(spin == 1.5 and isospin == 0.0 and Ns == 1))
        * pow(1.7302, gate(spin == 1.5 and isospin == 0.0 and Ns == 2))
        * pow(1.8640, gate(spin == 1.5 and isospin == 0.0 and Ns == 3))
        * pow(1.5664, gate(spin == 1.5 and isospin == 0.5) * Qs * Qc)
        * pow(1.6317, gate(spin == 1.5 and isospin == 1.0) * Qc)
        * pow(1.2837, gate(spin == 1.5 and isospin == 0.0) * Qb)
        * pow(2.0551, gate(spin == 1.5 and isospin == 1.0) * Qb)
        * pow(1.6290, gate(spin == 2.5))
    )
elif state == 3:
    stateExcitation = (
        E_debase
        * pow(1.6139, gate(spin == 0.5 and isospin == 0.0) * Qs)
        * pow(1.0885, gate(spin == 0.5 and Ns == 2) * Qc)
        * pow(1.7570, gate(spin == 0.5 and isospin == 0.0) * Qd * Qc)
        * pow(1.6749, gate(spin == 0.5 and isospin == 0.5))
        * pow(1.7878, gate(spin == 1.5 and Ns == 2) * Qc)
        * pow(1.7107, gate(spin == 1.5 and isospin == 0.0) * Qd * Qs)
        * pow(1.743, gate(spin == 1.5 and isospin == 0.0) * Qd * Qc)
        * pow(1.7205, gate(spin == 1.5 and isospin == 0.5) * Qs * Qc)
        * pow(2.0705, Qb)
        * pow(1.7572, gate(spin == 1.5 and isospin == 0.5 and Ns == 2))
        * pow(2.1219, gate(spin == 2.5) * Qs)
        * pow(1.7258, gate(spin == 2.5) * Qc)
    )
elif state == 4:
    stateExcitation = (
        E_debase
        * pow(1.8588, gate(spin == 0.5 and isospin == 0.0))
        * pow(1.8938, gate(spin == 0.5 and isospin == 0.5))
        * pow(1.8566, gate(spin == 2.5))
    )
elif state == 5:
    stateExcitation = E_debase * 2.0116

return stateExcitation

def nQuarks(quarks: list[str]) -> list[float]:
    total = 0.0
    Q = [0.0, 0.0, 0.0, 0.0, 0.0, 0.0]
    for q in quarks:
        if q == "u":
            Q[0] += 1.0
        elif q == "d":
            Q[1] += 1.0
        elif q == "s":
            Q[2] += 1.0
        elif q == "c":
            Q[3] += 1.0
        elif q == "b":
            Q[4] += 1.0
        elif q == "t":
            Q[5] += 1.0
        else:
            total += 0
    return Q

```

```

def predictedMass(hadron: Hadron) -> tuple[float, float]:
    q1 = hadron.quarks[0]
    q2 = hadron.quarks[1] if len(hadron.quarks) > 1 else "u"
    q3 = hadron.quarks[2] if len(hadron.quarks) > 2 else "u"
    G1 = generation.get(q1, 1.0)
    G2 = generation.get(q2, 1.0)
    G3 = generation.get(q3, 1.0)
    Q = float(len(hadron.quarks))
    G = float(max(G1, G2))

    Qu = 1.0 if "u" in hadron.quarks else 0.0
    Qd = 1.0 if "d" in hadron.quarks else 0.0
    Qs = 1.0 if "s" in hadron.quarks else 0.0
    Qc = 1.0 if "c" in hadron.quarks else 0.0
    Qb = 1.0 if "b" in hadron.quarks else 0.0

    NQ = nQuarks(hadron.quarks)
    Ns = NQ[2]
    Nc = NQ[3]
    Nb = NQ[4]

    G = max(G, float(G3))

    if Q == 1:
        delta_e = deltaEbase(hadron.quarks)
        return E_quark, delta_e
    else:
        a = B_base / E_base
        E_base_a = E_base * pow(a, Q - 2)
        delta_e = deltaEbase(hadron.quarks)
        state_excitation = stateEnergy(
            E_debase=E_base_a,
            state=float(hadron.state),
            spin=hadron.spin,
            isospin=hadron.isospin,
            isospin3=hadron.isospin3,
            sdsymmetry=hadron.sdsymmetry,
            G=float(G),
            Q=Q,
            Qu=Qu,
            Qd=Qd,
            Qs=Qs,
            Qc=Qc,
            Qb=Qb,
            Ns=Ns,
            Nc=Nc,
            Nb=Nb,
            orbital=hadron.orbital,
        )
        return state_excitation, delta_e + Q * E_quark

if __name__ == "__main__":
    totalError = 0.0

    print(len(hadrons))

    for had in hadrons:
        state_excitation, delta_eqe_quark = predictedMass(had)
        pred = state_excitation + delta_eqe_quark
        err = ((pred - had.mass) / had.mass) * 100.0
        correction = (had.mass - delta_eqe_quark) / state_excitation

        print(
            f"{had.name:<16} | Obs: {had.mass:9.5f} | Pred: {pred:9.5f} | Δ%: {err:+6.5f}"
        )

```

```

totalError += abs(err)

print(f"Average  $\Delta\%$  error: {totalError / float(len(hadrons)):.5f}%")

```

B Python Code Results

Table 2 and Table 3 list the observed masses, predicted masses, and percent deviations for the hadrons included in the benchmark set. The predicted values are generated directly from the explicit implementation of the mass equation provided in the appendix code listing.

Table 1: Quark benchmark results for the hadron mass equation.

Particle	Observed (MeV)	Predicted (MeV)	$\Delta\%$
u	2.17000	2.17000	+0.00000
d	4.67000	4.67000	+0.00000
s	93.40000	93.40000	+0.00000
c	1271.00000	1271.00000	+0.00000
b	4180.00000	4180.00000	+0.00000
t	172500.00000	172500.00000	+0.00000

Table 2: Meson benchmark results for the hadron mass equation.

Particle	Observed (MeV)	Predicted (MeV)	$\Delta\%$
Pi0	134.97660	134.97660	+0.00000
Pi+	139.57039	139.56389	-0.00466
Pi-	139.57039	139.56389	-0.00466
K+	493.67700	495.09992	+0.28823
K-	493.67700	495.09992	+0.28823
K0	497.61100	497.59992	-0.00223
K0	497.61100	497.59992	-0.00223
K0_S	497.61100	497.59992	-0.00223
K0_L	497.61100	497.59992	-0.00223
Eta	547.86200	547.85835	-0.00067
Rho0	775.26000	775.10861	-0.01953
Rho+	775.11000	775.10861	-0.00018
Rho-	775.11000	775.10861	-0.00018
Omega	782.66000	775.10861	-0.96484
K*+	891.67000	891.66988	-0.00001
K*-	891.67000	891.66988	-0.00001
K*0	895.55000	894.16988	-0.15411
K*0bar	895.55000	894.16988	-0.15411
EtaPrime	957.78000	957.77694	-0.00032
a0(980)+	980.00000	979.99904	-0.00010
a0(980)0	980.00000	979.99904	-0.00010
a0(980)-	980.00000	979.99904	-0.00010
Phi	1019.46100	1019.45725	-0.00037
f2(1270)	1275.50000	1275.49485	-0.00040
Eta(1295)	1294.00000	1293.99777	-0.00017
a2(1320)+	1318.20000	1318.18996	-0.00076

Particle	Observed (MeV)	Predicted (MeV)	$\Delta\%$
a2(1320)0	1318.20000	1318.18996	-0.00076
a2(1320)-	1318.20000	1318.18996	-0.00076
Omega(1420)	1410.00000	1409.99984	-0.00001
Rho(1450)+	1465.00000	1468.84017	+0.26213
Rho(1450)0	1465.00000	1468.84017	+0.26213
Rho(1450)-	1465.00000	1468.84017	+0.26213
a0(1450)+	1474.00000	1468.84017	-0.35006
a0(1450)0	1474.00000	1468.84017	-0.35006
a0(1450)-	1474.00000	1468.84017	-0.35006
Omega(1650)	1670.00000	1669.98900	-0.00066
Phi(1680)	1680.00000	1679.97580	-0.00144
Rho3(1690)+	1688.80000	1688.79945	-0.00003
Rho3(1690)0	1688.80000	1688.79945	-0.00003
Rho3(1690)-	1688.80000	1688.79945	-0.00003
Rho(1700)+	1720.00000	1719.98790	-0.00070
Rho(1700)0	1720.00000	1719.98790	-0.00070
Rho(1700)-	1720.00000	1719.98790	-0.00070
Eta(1760)	1751.00000	1750.99696	-0.00017
Pi(1800)+	1812.00000	1812.00279	+0.00015
Pi(1800)0	1812.00000	1812.00279	+0.00015
Pi(1800)-	1812.00000	1812.00279	+0.00015
Phi3(1850)	1854.00000	1853.99812	-0.00010
D0	1864.84100	1866.12212	+0.06870
D0	1864.84100	1866.12212	+0.06870
D+	1869.65000	1868.62212	-0.05498
D-	1869.65000	1868.62212	-0.05498
Ds+	1968.34000	1968.32525	-0.00075
Ds-	1968.34000	1968.32525	-0.00075
D*0	2006.85000	2006.84173	-0.00041
D*0bar	2006.85000	2006.84173	-0.00041
D*+	2010.26000	2009.34173	-0.04568
D*-	2010.26000	2009.34173	-0.04568
Ds*+	2112.20000	2112.19238	-0.00036
Ds*-	2112.20000	2112.19238	-0.00036
Eta(2225)	2221.00000	2220.99415	-0.00026
Ds0*(2317)+	2317.80000	2317.78756	-0.00054
Ds0*(2317)-	2317.80000	2317.78756	-0.00054
D0*(2300)0	2318.00000	2317.99584	-0.00018
D0*(2300)0bar	2318.00000	2317.99584	-0.00018
D0*(2400)+	2403.00000	2402.99018	-0.00041
D0*(2400)-	2403.00000	2402.99018	-0.00041
D1(2420)0	2420.10000	2423.45226	+0.13852
D1(2420)0bar	2420.10000	2423.45226	+0.13852
D1(2420)+	2423.40000	2425.95226	+0.10532
D1(2420)-	2423.40000	2425.95226	+0.10532
D1(2430)0	2427.00000	2423.45226	-0.14618
D1(2430)0bar	2427.00000	2423.45226	-0.14618
D2*(2460)0	2462.60000	2462.59799	-0.00008
D2*(2460)0bar	2462.60000	2462.59799	-0.00008
D2*(2460)+	2465.40000	2465.09799	-0.01225

Particle	Observed (MeV)	Predicted (MeV)	$\Delta\%$
D2*(2460)-	2465.40000	2465.09799	-0.01225
Ds1(2460)+	2459.50000	2496.65344	+1.51061
Ds1(2460)-	2459.50000	2496.65344	+1.51061
Ds1(2536)+	2535.10000	2496.65344	-1.51657
Ds1(2536)-	2535.10000	2496.65344	-1.51657
Ds2*(2573)+	2572.60000	2572.59872	-0.00005
Ds2*(2573)-	2572.60000	2572.59872	-0.00005
Ds0(2590)+	2591.00000	2590.98760	-0.00048
Ds0(2590)-	2591.00000	2590.98760	-0.00048
D*(2600)0	2628.00000	2626.77942	-0.04645
D*(2600)0bar	2628.00000	2626.77942	-0.04645
D*(2600)+	2628.00000	2629.27942	+0.04868
D*(2600)-	2628.00000	2629.27942	+0.04868
Ds1*(2700)+	2708.00000	2719.22928	+0.41467
Ds1*(2700)-	2708.00000	2719.22928	+0.41467
D*(2750)0	2763.50000	2763.50117	+0.00004
D*(2750)0bar	2763.50000	2763.50117	+0.00004
Ds1*(2860)+	2859.00000	2859.63880	+0.02234
Ds1*(2860)-	2859.00000	2859.63880	+0.02234
Ds3*(2860)+	2860.50000	2859.63880	-0.03011
Ds3*(2860)-	2860.50000	2859.63880	-0.03011
Eta_c(1S)	2983.90000	2983.89188	-0.00027
DsJ(3040)+	3044.00000	3043.98893	-0.00036
DsJ(3040)-	3044.00000	3043.98893	-0.00036
J/Psi	3096.90000	3096.89555	-0.00014
Chi_c0(1P)	3414.71000	3414.71276	+0.00008
Chi_c1(1P)	3510.67000	3525.37152	+0.41877
h_c(1P)	3525.38000	3525.37152	-0.00024
Chi_c2(1P)	3556.17000	3556.16275	-0.00020
Eta_c(2S)	3637.50000	3637.44230	-0.00159
Psi(2S)	3686.10000	3686.09326	-0.00018
Psi(3770)	3778.10000	3778.09534	-0.00012
Psi2(3823)	3822.20000	3822.19996	-0.00000
Psi3(3842)	3842.70000	3842.68900	-0.00029
Psi(4040)	4039.00000	4038.99427	-0.00014
Psi(4160)	4191.00000	4190.98991	-0.00024
Psi(4415)	4421.00000	4420.99510	-0.00011
B+	5279.10000	5279.09618	-0.00007
B-	5279.10000	5279.09618	-0.00007
B0	5279.10000	5281.59618	+0.04728
B0	5279.10000	5281.59618	+0.04728
B*+	5324.71000	5324.69999	-0.00019
B*-	5324.71000	5324.69999	-0.00019
Bs0	5366.88000	5366.87930	-0.00001
Bs0	5366.88000	5366.87930	-0.00001
Bs*0	5415.40000	5415.39182	-0.00015
Bs*0bar	5415.40000	5415.39182	-0.00015
B1(5721)+	5726.10000	5724.78090	-0.02304
B1(5721)-	5726.10000	5724.78090	-0.02304
B1(5721)0	5726.00000	5727.28090	+0.02237

Particle	Observed (MeV)	Predicted (MeV)	$\Delta\%$
B1(5721)0bar	5726.00000	5727.28090	+0.02237
B2*(5747)+	5739.50000	5724.78090	-0.25645
B2*(5747)-	5739.50000	5724.78090	-0.25645
B2*(5747)0	5739.50000	5727.28090	-0.21289
B2*(5747)0bar	5739.50000	5727.28090	-0.21289
Bs1(5830)0	5828.70000	5834.07816	+0.09227
Bs1(5830)0bar	5828.70000	5834.07816	+0.09227
Bs2*(5840)0	5839.90000	5834.07816	-0.09969
Bs2*(5840)0bar	5839.90000	5834.07816	-0.09969
BsJ(5850)0	5850.00000	5849.99273	-0.00012
BsJ(5850)0bar	5850.00000	5849.99273	-0.00012
BJ(5840)0	5860.80000	5862.72129	+0.03278
BJ(5840)0bar	5860.80000	5862.72129	+0.03278
BJ(5840)+	5862.90000	5860.22129	-0.04569
BJ(5840)-	5862.90000	5860.22129	-0.04569
BJ(5970)+	5961.00000	5961.00072	+0.00001
BJ(5970)-	5961.00000	5961.00072	+0.00001
BJ(5970)0	5971.00000	5970.99671	-0.00006
BJ(5970)0bar	5971.00000	5970.99671	-0.00006
BsJ(6063)0	6063.00000	6062.99420	-0.00010
BsJ(6063)0bar	6063.00000	6062.99420	-0.00010
BsJ(6114)0	6114.00000	6113.99763	-0.00004
BsJ(6114)0bar	6114.00000	6113.99763	-0.00004
Bc+	6274.90000	6274.89271	-0.00012
Bc-	6274.90000	6274.89271	-0.00012
Bc(2S)*+	6875.00000	6874.99485	-0.00007
Bc(2S)*-	6875.00000	6874.99485	-0.00007
Eta_b(1S)	9399.00000	9398.99562	-0.00005
Upsilon(1S)	9460.30000	9460.29617	-0.00004
Chi_b0(1P)	9859.40000	9892.79564	+0.33872
Chi_b1(1P)	9892.80000	9892.79564	-0.00004
h_b(1P)	9899.30000	9892.79564	-0.06571
Chi_b2(1P)	9912.20000	9892.79564	-0.19576
Eta_b(2S)	9999.00000	9998.99525	-0.00005
Upsilon(2S)	10023.26000	10023.25151	-0.00008
Chi_b0(2P)	10232.50000	10232.49858	-0.00001
Chi_b1(2P)	10255.50000	10259.75323	+0.04147
h_b(2P)	10259.80000	10259.75323	-0.00046
Chi_b2(2P)	10268.70000	10268.69717	-0.00003
Upsilon(3S)	10355.20000	10421.48725	+0.64013
Chi_b0(3P)	10495.00000	10421.48725	-0.70045
Chi_b1(3P)	10513.40000	10421.48725	-0.87424
Chi_b2(3P)	10524.00000	10421.48725	-0.97409
Upsilon(4S)	10579.40000	10579.40279	+0.00003
Upsilon(5S)	10885.00000	10884.99577	-0.00004
Upsilon(6S)	11000.00000	10999.99837	-0.00001

Table 3: Baryon benchmark results for the hadron mass equation.

Particle	Observed (MeV)	Predicted (MeV)	$\Delta\%$
Proton	938.27209	938.27209	+0.00000
Neutron	939.56542	939.56405	-0.00015
Lambda0	1115.68300	1115.92346	+0.02155
Sigma+	1189.37000	1190.08759	+0.06033
Sigma0	1192.64200	1192.58759	-0.00456
Sigma	1197.44900	1195.08759	-0.19720
Delta++	1232.00000	1226.90990	-0.41316
Delta+	1232.00000	1229.40990	-0.21024
Delta0	1232.00000	1231.90990	-0.00731
Delta	1232.00000	1234.40990	+0.19561
Xi0	1314.86000	1316.62955	+0.13458
Xi	1321.71000	1319.12955	-0.19524
Sigma*+	1382.80000	1381.51558	-0.09289
Sigma*0	1383.70000	1384.01558	+0.02281
Sigma*-	1387.20000	1386.51558	-0.04934
Lambda(1405)0	1405.10000	1405.10983	+0.00070
Lambda(1520)0	1519.50000	1519.50199	+0.00013
Xi*0	1531.80000	1532.12542	+0.02124
Xi*-	1535.00000	1534.62542	-0.02440
Lambda(1600)0	1600.00000	1599.97609	-0.00149
Omega	1672.45000	1672.42046	-0.00177
Lambda(1690)0	1690.00000	1689.92866	-0.00422
Lambda(1820)0	1820.00000	1825.50799	+0.30264
Xi(1690)0	1690.00000	1688.42731	-0.09306
Xi(1690)-	1690.00000	1690.92731	+0.05487
Xi(1820)0	1823.00000	1821.86934	-0.06202
Xi(1820)-	1823.00000	1824.36934	+0.07511
Xi(1950)0	1950.00000	1948.80654	-0.06120
Xi(1950)-	1950.00000	1951.30654	+0.06700
Omega(2012)-	2012.40000	2012.34453	-0.00276
Omega(2250)-	2252.00000	2252.00123	+0.00005
Lambda+c	2286.46000	2286.46107	+0.00005
Sigma++c	2453.97000	2450.39191	-0.14581
Sigma+c	2452.90000	2452.89191	-0.00033
Sigma0c	2453.75000	2455.39191	+0.06691
Xi0c	2470.85000	2470.61728	-0.00942
Xi+c	2468.00000	2468.11728	+0.00475
Sigma*++c	2518.40000	2514.97563	-0.13597
Sigma*+c	2517.50000	2517.47563	-0.00097
Sigma*0c	2518.10000	2519.97563	+0.07449
Xi'+c	2578.40000	2577.39077	-0.03914
Xi'0c	2578.80000	2579.89077	+0.04230
Lambda(2595)+c	2592.25000	2592.32168	+0.00277
Lambda(2625)+c	2628.11000	2628.03473	-0.00286
Xi*(2645)+c	2645.50000	2643.74782	-0.06623
Xi*(2645)0c	2646.30000	2646.24782	-0.00197
Omega0c	2695.20000	2695.20540	+0.00020
Omega*c	2765.90000	2765.92224	+0.00080

Particle	Observed (MeV)	Predicted (MeV)	$\Delta\%$
Xi(2790)+c	2791.90000	2791.59341	-0.01098
Xi(2790)0c	2793.90000	2794.09341	+0.00692
Sigma(2800)+c	2792.00000	2794.11695	+0.07582
Sigma(2800)++c	2801.00000	2791.61695	-0.33499
Sigma(2800)0c	2806.00000	2796.61695	-0.33439
Xi(2815)+c	2816.60000	2822.16614	+0.19762
Xi(2815)0c	2819.85000	2824.66614	+0.17079
Lambda(2860)+c	2856.10000	2897.54382	+1.45106
Lambda(2880)+c	2881.60000	2881.56051	-0.00137
Xi(2882)+c	2882.00000	2880.33794	-0.05767
Xi(2882)0c	2882.00000	2882.83794	+0.02908
Lambda(2910)+c	2910.60000	2910.55349	-0.00160
Lambda(2940)+c	2939.60000	2897.54382	-1.43068
Xi(2923)+c	2923.00000	2922.99107	-0.00031
Xi(2923)0c	2923.00000	2925.49107	+0.08522
Xi(2930)+c	2930.00000	2922.99107	-0.23921
Xi(2930)0c	2930.00000	2925.49107	-0.15389
Xi(2970)+c	2967.00000	2965.36542	-0.05509
Xi(2970)0c	2967.00000	2967.86542	+0.02917
Xi(3055)+c	3055.90000	3091.83799	+1.17602
Xi(3080)+c	3077.20000	3091.83799	+0.47569
Omega(3000)0c	3000.40000	3028.25293	+0.92831
Omega(3050)0c	3050.20000	3028.25293	-0.71953
Omega(3065)0c	3065.60000	3065.60927	+0.00030
Omega(3090)0c	3090.20000	3090.26273	+0.00203
Omega(3120)0c	3119.10000	3119.13476	+0.00111
Xi(3123)+c	3122.90000	3091.83799	-0.99465
Xi(3123)0c	3122.90000	3094.33799	-0.91460
Omega(3185)0c	3185.10000	3185.11237	+0.00039
Omega(3327)0c	3327.10000	3327.10362	+0.00011
Sigma++cc	3621.70000	3626.76033	+0.13972
Sigma+cc	3619.97000	3629.26033	+0.25664
Omega+cc	3730.97000	3717.99033	-0.34789
Lambda0b	5619.60000	5619.44751	-0.00271
Xi0b	5787.80000	5788.47066	+0.01159
Xib	5794.40000	5790.97066	-0.05918
Sigma+b	5810.55000	5829.69145	+0.32943
Sigma0b	5813.40000	5832.19145	+0.32324
Sigma-b	5815.50000	5834.69145	+0.33001
Sigma*+b	5830.30000	5829.69145	-0.01044
Sigma*0b	5831.80000	5832.19145	+0.00671
Sigma*-b	5834.70000	5834.69145	-0.00015
Lambda(5912)0b	5912.20000	5912.10799	-0.00156
Lambda(5920)0b	5920.10000	5920.08498	-0.00025
Xi'0b	5935.90000	5934.01171	-0.03181
Xi'-b	5935.00000	5936.51171	+0.02547
Xi(5945)0b	5945.30000	5920.92145	-0.41005
Xi*(5955)-b	5955.30000	5923.42145	-0.53530
Omegab	6045.10000	6045.09970	-0.00000
Omega*b	6045.20000	6012.15145	-0.54669

Particle	Observed (MeV)	Predicted (MeV)	$\Delta\%$
Lambda(6070)0b	6072.00000	6110.87716	+0.64027
Sigma(6097)+b	6095.80000	6094.06652	-0.02844
Sigma(6097)-b	6098.00000	6099.06652	+0.01749
Lambda(6146)0b	6146.20000	6110.87716	-0.57471
Lambda(6152)0b	6152.50000	6110.87716	-0.67652

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